

Production of quarkonium pairs via the fragmentation mechanism

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In this manuscript, we analyze the contribution of fragmentation mechanisms to the inclusive hadroproduction of heavy quarkonium pairs, taking into account both single- and double-hadron (dihadron) fragmentation channels. To estimate the contribution of the latter mechanism, we construct a microscopic perturbative model of the dihadron fragmentation function, which is valid in the limit of large invariant masses of the heavy quarkonium pair. Using the Color Glass Condensate framework to evaluate the $Q\bar{Q}$ production amplitude, we find that in LHC kinematics, the single fragmentation of heavy quarks into quarkonium is only a minor correction, whereas dihadron fragmentation can provide a sizable contribution, on par with the contributions of single- and double-parton scattering.

I. INTRODUCTION

Due to their large masses, heavy quarkonia are widely used as important probes of the gluonic field of the target [1, 2]. The nonrelativistic QCD (NRQCD) framework allows us to systematically evaluate various perturbative corrections to processes involving quarkonium [3–12]. While the production of single quarkonium is understood reasonably well [13–17], due to the large number of contributing mechanisms the information about the target which can be extracted from these data is limited. Furthermore, the corresponding cross sections frequently involve unknown NRQCD long-distance matrix elements (LDMEs) that must be extracted from the same processes, thus limiting the predictive power of the analysis and reducing it to a mere consistency check. For this reason, since the early days of QCD, theoretical efforts have been dedicated to the production mechanisms of multiple quarkonia in the final state [18–21], which can provide much more detailed information about the gluonic content of the target. Since the cross sections for multiple charmonium production decrease rapidly with the number of produced charmonia, current theoretical and experimental efforts focus primarily on the production of quarkonium pairs [22–26]. Available experimental data [27, 28] demonstrate that simple gluon-gluon fusion is insufficient to describe the inclusive hadroproduction cross section, signaling significant contributions from poorly known double-parton scattering (DPS) mechanisms. While DPS could potentially resolve the observed discrepancy, the required magnitude of DPS varies significantly across different channels: the value of the effective cross section σ_{eff} , which parameterizes the magnitude of double-parton scattering, varies by a factor of 2–3. This large uncertainty indicates that other production mechanisms may play an important role and warrant detailed analysis.

In this paper, we focus on the contribution of the fragmentation mechanism, which was introduced in the Collins-Soper formalism [29]. Although this mechanism typically dominates at very high energies, it is frequently overlooked in analyses of quarkonium pair production. The underlying assumption of the fragmentation picture is that the formation time of the final-state mesons (the fragmentation time) significantly exceeds the coherence (Ioffe) time t_c of $Q\bar{Q}$ pair formation [30, 31] as well as the shockwave interaction time with the target, t_h . The nonperturbative dynamics of the final-state hadronization are encoded in universal (hadron-dependent) fragmentation functions [32, 33]. Since the latter are essentially nonperturbative objects, at present most of their parametrizations rely on purely phenomenological extractions and carry sizable uncertainties. The situation is different for heavy quarkonium because the heavy-quark mass serves as a natural hard scale and justifies the use of a perturbative framework over a wide kinematic range [34–36]. In what follows, we distinguish between single- and double-hadron (dihadron) fragmentation, where a heavy quark Q fragments into one or two quarkonium states, respectively. The latter mechanism should not be confused with the so-called double-parton fragmentation discussed in Refs. [36–39], which assumes the fusion of a quark-antiquark pair into a single quarkonium state and thus contributes to the direct production mechanism.

The role of the fragmentation mechanism in inclusive quarkonium production has been discussed in Refs. [40, 41], and in the context of double quarkonium production in Refs. [42, 43]. However, those studies focused primarily on the single-fragmentation mechanism and were performed within the k_T -factorization framework, which is valid at relatively large values of x . In this manuscript, we focus on the role of double-quarkonium fragmentation and its contribution to inclusive quarkonium pair production in high-energy pp collisions at the LHC. To account for the potential onset of gluon saturation effects at these energies, we use the Color Glass Condensate (CGC) framework [44–53], which naturally incorporates saturation effects and provides a phenomenologically successful description of both hadron-hadron and lepton-hadron collisions [54–60]. Previous studies of quarkonium pairs in the CGC framework have either focused on direct quarkonium pair production [61, 62] or analyzed pair production combining the CGC with the color evaporation model [63], which can be viewed as a special case when the fragmentation function is

constant.

The paper is structured as follows. In Sec. II, we formulate the theoretical framework used to evaluate the contributions of the fragmentation mechanism. In Sec. II A, we briefly summarize the CGC framework used to describe the interaction of high-energy partons with the target. In Sec. II B, we analytically evaluate the dihadron fragmentation functions (DiFFs) in the heavy-quark mass limit within the NRQCD framework and discuss their properties (technical details and final expressions are provided in Appendix A). In Sec. II C, we demonstrate that within the CGC picture, the cross section of the fragmentation mechanism can be naturally expressed as a convolution of the fragmentation functions and dipole production cross sections, showing that no additional kinematic cuts on transverse momenta are required for applicability of the heavy-quark fragmentation. In Sec. III, we present our numerical estimates for the contributions of the above-mentioned fragmentation mechanisms to S -wave charmonium pairs and compare the predicted results with experimental data. Finally, we draw our conclusions in Sec. IV.

II. THEORETICAL FRAMEWORK

High-energy hadronic collisions at the LHC have been used to study gluon-mediated hadroproduction and photon-mediated production in ultraperipheral kinematics. In what follows, we focus on the hadroproduction of heavy quarkonium pairs in forward kinematics, where the process can be described in the dilute-dense approximation. The cross section can be written as

$$\frac{d\sigma_{pp \rightarrow \mathcal{Q}_1 \mathcal{Q}_2 X}}{dy_1 d^2 \mathbf{k}_1^\perp dy_2 d^2 \mathbf{k}_2^\perp} \approx \int dx_1 d^2 \mathbf{q}_\perp \mathcal{F}(x_1, \mathbf{q}_\perp, \mu^2) \left. \frac{d\sigma(g + p \rightarrow \mathcal{Q}_1 + \mathcal{Q}_2 + X)}{dy_1 d^2 \mathbf{p}_1^\perp dy_2 d^2 \mathbf{p}_2^\perp} \right|_{\mathbf{p}_a^\perp = \mathbf{k}_a^\perp - \beta_a \mathbf{q}_\perp}, \quad (1)$$

where $(y_a, \mathbf{k}_a^\perp)$ are the rapidities and transverse momenta of the produced quarkonia. The variables $\beta_a = p_a^+/q^+$ are the fractions of the incoming gluon's momentum carried by the quarkonia, and the momenta $\mathbf{p}_a^\perp = \mathbf{k}_a^\perp - \beta_a \mathbf{q}_\perp$ correspond to their transverse momenta with respect to the incoming gluon. The function $\mathcal{F}(x_1, \mathbf{q}_\perp, \mu^2)$ is the transverse-momentum-dependent parton distribution function (TMD PDF) of the gluon. At very large transverse momenta, we can approximate $\mathbf{p}_a^\perp = \mathbf{k}_a^\perp - z_a \mathbf{q}_\perp \approx \mathbf{k}_a^\perp$ and reduce Eq. (1) to

$$\frac{d\sigma_{pp \rightarrow \mathcal{Q}_1 \mathcal{Q}_2 X}}{dy_1 d^2 \mathbf{k}_1^\perp dy_2 d^2 \mathbf{k}_2^\perp} \approx \int dx_1 x_1 g(x_1, \mu^2) \left. \frac{d\sigma(g + p \rightarrow \mathcal{Q}_1 + \mathcal{Q}_2 + X)}{dy_1 d^2 \mathbf{p}_1^\perp dy_2 d^2 \mathbf{p}_2^\perp} \right|_{\mathbf{p}_a^\perp \approx \mathbf{k}_a^\perp}, \quad (2)$$

where we introduced the collinear gluon PDF $x_1 g(x_1, \mu^2) = \int d^2 \mathbf{q}_\perp \mathcal{F}(x_1, \mathbf{q}_\perp, \mu^2)$, in agreement with the findings of Refs. [64–67]. In essence, the collinear approximation (2) reflects the fact that the typical transverse momentum of a gluon inside the proton is significantly smaller than the invariant mass of the quarkonium pair. Alternatively, if we integrate both sides of Eq. (1) over $\mathbf{k}_1^\perp$ and $\mathbf{k}_2^\perp$, and shift the integration variables on the right-hand side as $\mathbf{k}_a^\perp \rightarrow \mathbf{p}_a^\perp + z_a \mathbf{q}_\perp$, we can separate the integrals over $\mathbf{q}_\perp$, $\mathbf{p}_1^\perp$, and $\mathbf{p}_2^\perp$ to obtain the transverse-momentum-integrated analog of Eq. (1):

$$\frac{d\sigma_{pp \rightarrow \mathcal{Q}_1 \mathcal{Q}_2 X}}{dy_1 dy_2} \approx \int dx_1 x_1 g(x_1, \mu^2) \frac{d\sigma(g + p \rightarrow \mathcal{Q}_1 + \mathcal{Q}_2 + X)}{dy_1 dy_2}. \quad (3)$$

As we will see below, for heavy quarks of mass m_Q produced in forward kinematics, the variable x_1 remains relatively large ($x_1 \gtrsim 10^{-1}$), meaning that the dilute-dense approximation remains valid even when integrating down to the region of small $k_\perp \sim Q_s(x)$, where $Q_s(x) \lesssim m_Q$ is the saturation scale. For light quarks with mass $m_q \ll Q_s(x)$, the relation (3) does not hold, and the collinear approximation is valid only in the region of large transverse momenta.

To proceed with the evaluations, we work in the reference frame, where the colliding protons have the same energies E_p but move in opposite directions along axis z . In what follows, we write out the explicit light-cone decompositions of the participant momenta. We denote the momenta of the colliding hadrons (the dilute projectile and the dense target) as P_1 and P_2 , the momentum of the gluon emitted from the projectile as q , and the four-momenta of the produced heavy quarkonia as p_1 and p_2 . The light-cone expansion of these momenta is given by

$$q = (q^+, 0, \mathbf{q}_\perp), \quad q^+ \approx x_1 P_1^+ \quad (4)$$

$$P_1 = \left(P^+, \frac{m_N^2}{2P^+}, \mathbf{0}_\perp \right), \quad P_2 = \left(\frac{m_N^2}{2P^+}, P^+, \mathbf{0}_\perp \right), \quad P^+ = \frac{E_p + \sqrt{E_p^2 - m_N^2}}{\sqrt{2}} \approx \sqrt{2} E_p \quad (5)$$

$$p_a = \left(\frac{M_a^\perp e^{y_a}}{\sqrt{2}}, \frac{M_a^\perp e^{-y_a}}{\sqrt{2}}, \mathbf{p}_a^\perp \right), \quad M_a^\perp \equiv \sqrt{M_a^2 + (\mathbf{p}_a^\perp)^2}, \quad a = 1, 2, \quad (6)$$

where m_N is the nucleon mass and M_1, M_2 are the masses of the produced quarkonia. The proton energy E_p in symmetric kinematics is related to the Mandelstam invariant $s = (P_1 + P_2)^2$ by $E_p = \sqrt{s}/2$. In what follows, we are primarily interested in the high-energy collider kinematics $q^+, P^+ \gg \{M_a, m_N\}$, where the quarkonia are produced with relatively small transverse momenta. Since the plus-component of the light-cone momentum of any particle is always positive, for the inclusive production of the quarkonium pair the variable q^+ is kinematically bounded by

$$q^+ \gtrsim \frac{M_1^\perp e^{y_1} + M_2^\perp e^{y_2}}{\sqrt{2}}. \quad (7)$$

The invariant mass M_{12} of the produced heavy quarkonium pair is given in terms of these variables by

$$M_{12}^2 = (p_1 + p_2)^2 = M_1^2 + M_2^2 + 2(M_1^\perp M_2^\perp \cosh \Delta y - \mathbf{p}_1^\perp \cdot \mathbf{p}_2^\perp), \quad \Delta y = y_1 - y_2, \quad (8)$$

which, along with the heavy quarkonium masses, serves as a hard scale justifying the perturbative analysis of the dihadron fragmentation function. The invariant mass M_{12} also can be used as a convenient indicator that the produced quarkonia are well separated kinematically. To suppress the contribution of the near-threshold region in different observables, which may receive sizable corrections from soft final-state interactions between the produced quarkonia, we will sometimes present results by introducing a lower cut on the invariant mass, $M_{12} > M_1 + M_2 + \Lambda$ with $\Lambda \approx 1$ GeV, or equivalently, on the relative velocity:

$$v_{\text{rel}} = \sqrt{1 - \frac{p_1^2 p_2^2}{(p_1 \cdot p_2)^2}} = \sqrt{1 - \frac{4M_1^2 M_2^2}{(M_{12}^2 - M_1^2 - M_2^2)^2}} \gtrsim 0.75 > 2\alpha_s(m_c). \quad (9)$$

This condition is automatically satisfied for any transverse momentum if the rapidity difference between the quarkonia is sufficiently large, $|\Delta y| > 0.8$. The cross section of the process $g + p \rightarrow Q_1 + Q_2 + X$, which appears on the right-hand side of Eqs. (1)–(3), is the central quantity of interest in this study and will be evaluated using the Color Glass Condensate approach [44–53] in Secs. II A and II C.

A. High-energy scattering in the CGC picture

The interaction of partons with an ultrarelativistic target (shockwave) in small- x kinematics is characterized by a nearly instantaneous color exchange. In the eikonal approximation, this can be described by neglecting the transverse motion of the parton and the change of its helicity during the interaction. In this picture, the interaction of heavy partons with the gluonic field of the target is described by Wilson lines $U(\mathbf{x}_\perp)$ defined as [44–53]

$$U(\mathbf{x}_\perp) = \hat{P} \exp \left(ig \int_{-\infty}^{+\infty} dx^- A_a^+(x^-, \mathbf{x}_\perp) t^a \right), \quad (10)$$

where $\mathbf{x}_\perp$ is the impact parameter of the parton, t^a are the color generators in the corresponding representation (**3**, $\bar{\mathbf{3}}$, or **8** for a quark, antiquark, or gluon, respectively), $A_a^+(x)$ is the background gluonic field of the hadron, and operator $\hat{P}$ indicates path-ordering. Formally, this interaction can be represented via the CGC Feynman rules [68–71] formulated in terms of the effective vertices (T -matrix elements) describing high-energy parton scattering off the shockwave:

$$T_{\sigma, \sigma'; i, i'}^Q = 2\pi \delta(\ell_Q^+ - \ell'^+_Q) \gamma_{\sigma', \sigma}^+ \int d^2 \mathbf{z} e^{i(\ell_Q - \ell'_Q) \cdot \mathbf{z}} [U(\mathbf{z}) - 1]_{i', i} \quad (11)$$

$$T_{\sigma, \sigma'; i, i'}^{\bar{Q}} = -2\pi \delta(\ell_Q^+ - \ell'^+_Q) \gamma_{\sigma, \sigma'}^+ \int d^2 \mathbf{z} e^{i(\ell_{\bar{Q}} - \ell'_{\bar{Q}}) \cdot \mathbf{z}} [U^\dagger(\mathbf{z}) - 1]_{i, i'} \quad (12)$$

$$T_{\mu, \nu; a, b}^g = -2\pi \delta(\ell_g^+ - \ell'^+_g) g_{\mu\nu} \text{sgn}(\ell_g^+) \int d^2 \mathbf{z} e^{-i(\ell'_g - \ell_g) \cdot \mathbf{z}} [\mathcal{U}^{\text{sgn}(\ell_g^+)}(\mathbf{z}) - 1]_{b, a} \quad (13)$$

where ℓ_i and ℓ'_i ($i = Q, \bar{Q}, g$) are the parton momenta before and after the interaction, respectively; σ, σ' are Dirac indices; i, i' are color indices in the fundamental representation (**3** or $\bar{\mathbf{3}}$); and a, b are adjoint color indices. The matrices U and $\mathcal{U}$ represent the Wilson lines in the fundamental and adjoint representations, respectively. The factor $2\pi \delta(\ell_i^+ - \ell'^+_i)$ in the vertices reflects the conservation of longitudinal momentum in the eikonal approximation. In

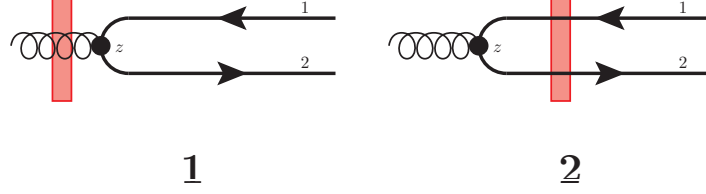


Figure 1. Diagrams contributing to inclusive heavy-quark pair hadroproduction in the CGC framework at leading order in α_s . The subscript z denotes the coordinate of the interaction vertex in configuration space, and the red block represents the shockwave interaction with the target.

the classical CGC framework [45–47], physical observables are averaged over the color charge configurations ρ in the target with a weight functional $W_Y[\rho]$. For many processes, this averaging can be expressed in terms of universal, process-independent multipole S -matrix elements [53, 57, 67]:

$$S_{2n}(Y, \mathbf{x}_1, \boldsymbol{\xi}_1, \dots, \mathbf{x}_n, \boldsymbol{\xi}_n) = \frac{1}{N_c} \langle \text{tr} (U(\mathbf{x}_1) U^\dagger(\boldsymbol{\xi}_1) \dots U(\mathbf{x}_n) U^\dagger(\boldsymbol{\xi}_n)) \rangle_Y, \quad (14)$$

which correspond to dipoles for $n = 1$, quadrupoles for $n = 2$, etc. For example, the cross section for heavy-quark pair production can be obtained by squaring the sum of the diagrams shown in Fig. 1, projecting the final-state quarks onto states with definite momenta and averaging over color sources using (14). The final result of this procedure yields (see Refs. [66, 69–71] for details)

$$\frac{d\sigma(g + p \rightarrow Q + \bar{Q} + X)}{dy_1 d^2\mathbf{p}_1^\perp dy_2 d^2\mathbf{p}_2^\perp} = \frac{1}{(2\pi)^4} \int d\alpha \int d^8\mathbf{X} \mathcal{R}_g(\alpha, \mathbf{r}'_{12}, \mathbf{r}_{12}) \Xi_{\text{LO}}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}'_1, \mathbf{x}'_2) \times \quad (15)$$

$$\times e^{-i(\mathbf{p}_{1\perp} + \mathbf{p}_{2\perp}) \cdot (\mathbf{b}_{12} - \mathbf{b}'_{12})} e^{-i(\bar{\alpha}\mathbf{p}_{1\perp} - \alpha\mathbf{p}_{2\perp}) \cdot (\mathbf{r}_{12} - \mathbf{r}'_{12})},$$

where α is the light-cone momentum fraction of the incoming gluon carried by the quark, $\mathbf{x}_1$ and $\mathbf{x}_2$ are the transverse coordinates of the quark and antiquark in the amplitude, while $\mathbf{x}'_1$ and $\mathbf{x}'_2$ are the corresponding variables in the conjugate amplitude. We define the phase-space volume element as $d^8\mathbf{X} = d^2\mathbf{x}_1 d^2\mathbf{x}_2 d^2\mathbf{x}'_1 d^2\mathbf{x}'_2$, and introduce the dipole sizes and impact parameters:

$$\mathbf{r}_{12} = \mathbf{x}_1 - \mathbf{x}_2, \quad \mathbf{r}'_{12} = \mathbf{x}'_1 - \mathbf{x}'_2, \quad \mathbf{b}_{12} = \alpha\mathbf{x}_1 + \bar{\alpha}\mathbf{x}_2, \quad \mathbf{b}'_{12} = \alpha\mathbf{x}'_1 + \bar{\alpha}\mathbf{x}'_2, \quad (16)$$

with $\bar{\alpha} \equiv 1 - \alpha$. The quantity $\mathcal{R}_g$ is the perturbative impact factor, given explicitly by

$$\mathcal{R}_g(\alpha, \mathbf{r}'_{12}, \mathbf{r}_{12}) = \frac{\alpha_s}{4\pi^2} \alpha(1 - \alpha) \left\{ m^2 K_1(mr_{12}) K_1(mr'_{12}) [e^{i\theta_{12}} \alpha^2 + e^{-i\theta_{12}} (1 - \alpha)^2] \right. \quad (17)$$

$$\left. + m^2 K_0(mr_{12}) K_0(mr'_{12}) \right\},$$

where m is the heavy-quark mass, α_s is the running coupling, and θ_{12} is the angle between the transverse vectors $\mathbf{r}_{12}$ and $\mathbf{r}'_{12}$. The physically relevant real part of the expression $[e^{i\theta_{12}} \alpha^2 + e^{-i\theta_{12}} (1 - \alpha)^2]$ can be rewritten as $[\alpha^2 + (1 - \alpha)^2] \cos \theta_{12} = [\alpha^2 + (1 - \alpha)^2] \hat{\mathbf{r}}_{12} \cdot \hat{\mathbf{r}}'_{12}$, in agreement with Refs. [70, 71]. The impact factor $\mathcal{R}_g$ can also be represented as a convolution of the gluon light-cone wave functions. The function Ξ_{LO} encodes the interaction of the partons with the target and is given by [66, 70, 71]

$$\Xi_{\text{LO}} = \frac{1}{N_c^2 - 1} \left[-S_4(Y, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}'_2, \mathbf{x}'_1) + N_c^2 S_2(Y, \mathbf{x}'_2, \mathbf{x}_2) S_2(Y, \mathbf{x}_1, \mathbf{x}'_1) - N_c^2 S_2(Y, \mathbf{x}_1, \mathbf{b}'_{12}) S_2(Y, \mathbf{b}'_{12}, \mathbf{x}_2) \right. \quad (18)$$

$$\left. - N_c^2 S_2(Y, \mathbf{x}'_1, \mathbf{b}_{12}) S_2(Y, \mathbf{b}_{12}, \mathbf{x}'_2) + S_2(Y, \mathbf{x}_1, \mathbf{x}_2) + S_2(Y, \mathbf{x}'_1, \mathbf{x}'_2) \right. \\ \left. + N_c^2 S_2(Y, \mathbf{b}_{12}, \mathbf{b}'_{12}) S_2(Y, \mathbf{b}'_{12}, \mathbf{b}_{12}) - 1 \right],$$

where we have taken into account that, in the eikonal picture, the transverse coordinate of the incoming gluon coincides with the impact parameter of the produced dipole [73]. Consequently, terms containing $\mathbf{b}_{12}$ or $\mathbf{b}'_{12}$ represent the interaction of the shockwave with the incoming gluon in the amplitude and the conjugate amplitude, respectively. The dipole S -matrix $S_2(\mathbf{x}_q, \mathbf{x}_{\bar{q}})$ is typically expressed in terms of the forward dipole scattering amplitude $N(Y, \mathbf{r}, \mathbf{b})$ as

$$N(Y, \mathbf{r}, \mathbf{b}) = 1 - S_2(Y, \mathbf{x}_q, \mathbf{x}_{\bar{q}}), \quad (19)$$

where $\mathbf{r} \equiv \mathbf{x}_q - \mathbf{x}_{\bar{q}}$ is the transverse size of the dipole, and $\mathbf{b} \equiv \alpha_q \mathbf{x}_q + \alpha_{\bar{q}} \mathbf{x}_{\bar{q}}$ is the impact parameter of its center of mass [73]. The quadrupole contribution $S_4(Y, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}'_2, \mathbf{x}'_1)$ is much less understood and phenomenologically constrained (see, however, model-dependent parameterizations in Refs. [74–76]). Fortunately, for the $\mathbf{p}_\perp$ -integrated cross section, this quadrupole uncertainty is irrelevant because the quadrupole contribution reduces to a dipole, namely

$$\begin{aligned} \frac{d\sigma(g + p \rightarrow Q + \bar{Q} + X)}{dy_1 dy_2} &= \int d^4 \mathbf{X} \mathcal{R}_g(\alpha, \mathbf{r}_{12}, \mathbf{r}_{12}) \overline{\Xi_{\text{LO}}(\mathbf{x}_1, \mathbf{x}_2)}, \\ d^4 \mathbf{X} &= d^2 \mathbf{x}_1 d^2 \mathbf{x}_2 = d^2 \mathbf{b}_{12} d^2 \mathbf{r}_{12}, \end{aligned} \quad (20)$$

where

$$\begin{aligned} \overline{\Xi_{\text{LO}}} &= -\frac{2}{N_c^2 - 1} [(1 - S_2(Y, \mathbf{x}_1, \mathbf{x}_2)) + N_c^2 [S_2(Y, \mathbf{x}_1, \mathbf{b}_{12}) S_2(Y, \mathbf{b}_{12}, \mathbf{x}_2) - 1]] = \\ &= -\frac{2N_c^2}{N_c^2 - 1} \left[\frac{1}{N_c^2} N(Y, \mathbf{r}_{12}, \mathbf{b}_{12}) + N(Y, \bar{\alpha} \mathbf{r}_{12}, \mathbf{b}_{1,12}) N(Y, -\alpha \mathbf{r}_{12}, \mathbf{b}_{2,12}) \right. \\ &\quad \left. - N(Y, \bar{\alpha} \mathbf{r}_{12}, \mathbf{b}_{1,12}) - N(Y, -\alpha \mathbf{r}_{12}, \mathbf{b}_{2,12}) \right] \end{aligned} \quad (21)$$

and in the last line we simplified the arguments of the dipole amplitude N using a set of identities

$$\mathbf{x}_1 - \mathbf{b}_{12} = \bar{\alpha}(\mathbf{x}_1 - \mathbf{x}_2), \quad \mathbf{x}_2 - \mathbf{b}_{12} = -\alpha(\mathbf{x}_1 - \mathbf{x}_2), \quad (22)$$

$$\mathbf{b}_{1,12} = \frac{\alpha \mathbf{x}_1 + \mathbf{b}_{12}}{1 + \alpha} = \mathbf{b}_{12} + \frac{\alpha \bar{\alpha} \mathbf{r}_{12}}{1 + \alpha}, \quad (23)$$

$$\mathbf{b}_{2,12} = \frac{\bar{\alpha} \mathbf{x}_2 + \mathbf{b}_{12}}{1 + \bar{\alpha}} = \mathbf{b}_{12} - \frac{\alpha \bar{\alpha} \mathbf{r}_{12}}{1 + \bar{\alpha}}. \quad (24)$$

To avoid the uncertainty associated with the quadrupole parameterization, we focus on the $\mathbf{p}_\perp$ -integrated cross sections in the remainder of this work.

B. Fragmentation function of the quarkonium

The concept of fragmentation functions was introduced in the Collins-Soper formalism [29]. These functions are essentially nonperturbative objects. Although they are generally frame-independent, a suitable choice of reference frame simplifies their formal definitions. For the single-quarkonium collinear fragmentation function, it is convenient to choose a frame where the final-state quarkonium has no transverse momentum, $\mathbf{p}_Q^\perp = \mathbf{0}_\perp$. In this frame, the leading-twist fragmentation function $D_1^{\mathcal{Q}/q}(z, \mu)$ is defined as [32, 33, 77]

$$\begin{aligned} D_1^{\mathcal{Q}/q}(z, \mu) &= \frac{z}{4} \sum_X \int \frac{d\xi^-}{2\pi} e^{ik^+ \xi^-} \text{Tr} \left[\langle 0 | \mathcal{W}(\infty^-, \xi^-) \psi_q(0^+, \xi^-, \mathbf{0}_T) | p_Q; X \rangle \right. \\ &\quad \left. \times \langle p_Q; X | \bar{\psi}_q(0^+, 0^-, \mathbf{0}_T) \mathcal{W}(0^-, \infty^-) | 0 \rangle \gamma^- \right], \end{aligned} \quad (25)$$

where ψ_q and $\bar{\psi}_q$ are the quark and antiquark field operators, and $\mathcal{W}$ are the usual gauge links defined as

$$\mathcal{W}(a^-, b^-) = \hat{P} \exp \left[ig \int_{(0, a, \mathbf{0}_\perp)}^{(0, b, \mathbf{0}_\perp)} d\zeta A_a^+(\zeta) t^a \right]. \quad (26)$$

The variable $z = p_Q^+/k^+$ is the light-cone momentum fraction of the parent quark carried by the quarkonium $\mathcal{Q}$. The dependence of the fragmentation function on the factorization scale μ is governed by the DGLAP evolution equations. In this reference frame, the transverse-momentum-dependent fragmentation function (TMD FF) $D_1^{\mathcal{Q}/q}(z, \mathbf{k}_\perp, \mu)$ is a straightforward generalization with a nonzero transverse separation and a subsequent Fourier transform over it, where $\mathbf{k}_\perp$ represents the transverse momentum of the quark prior to fragmentation.

For the unpolarized dihadron (diquarkonium) fragmentation function, the definition is simplest in the reference frame where the total momentum of the quarkonium pair $P_h^\mu = p_1^\mu + p_2^\mu$ does not have transverse component, namely $\mathbf{p}_{1\perp} + \mathbf{p}_{2\perp} = \mathbf{0}_\perp$, and the light-cone decomposition of the four-momenta of quarkonia is given by

$$p_a = \left(\frac{(1 \pm \zeta)}{2} P_h^+, \frac{M_a^2 + R_\perp^2}{(1 \pm \zeta) P_h^+}, \pm \mathbf{R}_\perp \right), \quad a = 1, 2; \quad \zeta = \frac{p_1^+ - p_2^+}{p_1^+ + p_2^+}. \quad (27)$$

The invariant mass of the produced quarkonia system in this frame is given by

$$M_{12}^2 = (p_1 + p_2)^2 = 2 \left(\frac{M_1^2}{1 + \zeta} + \frac{M_2^2}{1 - \zeta} \right) + \frac{4R_\perp^2}{1 - \zeta^2} \geq 2 \left(\frac{M_1^2}{1 + \zeta} + \frac{M_2^2}{1 - \zeta} \right), \quad (28)$$

and the dihadron fragmentation function is defined as [77]

$$D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta) = \frac{z}{4} \sum_X \int \frac{d\xi^-}{2\pi} e^{ik^+ \xi^-} \text{Tr} \left[\langle 0 | \mathcal{W}(\infty^-, \xi^-) \psi_q(0^+, \xi^-, \mathbf{0}_T) | P_1, P_2; X \rangle \right. \\ \left. \times \langle P_1, P_2; X | \bar{\psi}_q(0^+, 0^-, \mathbf{0}_T) \mathcal{W}(0^-, \infty^-) | 0 \rangle \gamma^- \right], \quad (29)$$

where z_1 and z_2 are the light-cone momentum fractions carried by the produced quarkonia, with $z = z_1 + z_2$, and

$$z_1 = \frac{p_1^+}{k^+} = z \frac{1 + \zeta}{2}, \quad z_2 = \frac{p_2^+}{k^+} = z \frac{1 - \zeta}{2}. \quad (30)$$

In general, both single and double fragmentation functions are nonperturbative and cannot currently be calculated from first principles. However, for heavy quarkonium, it is possible to justify a perturbative treatment in the heavy-quark mass limit and evaluate them analytically within the NRQCD framework. For the single S -wave quarkonium production, the corresponding fragmentation functions have been evaluated in Ref. [36] and are given by

$$D_1^{\mathcal{Q}/c}(z, \mu) \approx \alpha_s^2(\mu^2) \left[\frac{2}{3} \frac{C_F^2}{N_c} \frac{(z-1)^2}{(z-2)^6} z (5z^4 - 32z^3 + 72z^2 - 32z + 16) \right] \frac{\langle \mathcal{O}_{\mathcal{Q}}(^3S_1^{[1]}) \rangle}{m_c^3}, \quad (31)$$

$$D_1^{\mathcal{Q}/b}(z, \mu) \approx \frac{\alpha_s^2(\mu^2)}{12N_c z} \left[(z^2 - 2z + 2) \ln \left[\frac{\mu_0^2}{4m_c^2(1 - z + z^2\eta/4)} \right] - 2z^2 \left(1 + \frac{1 - z - z^2/2}{4 - 4z + z^2\eta} \right) \right] \frac{\langle \mathcal{O}_{\mathcal{Q}}(^3S_1^{[8]}) \rangle}{m_c^3} \quad (32)$$

where $\eta = m_b/m_c$, μ_0 is the initial factorization scale at which the NRQCD matrix elements are defined, and $\langle \mathcal{O}_{\mathcal{Q}}(^3S_1^{[1]}) \rangle$, $\langle \mathcal{O}_{\mathcal{Q}}(^3S_1^{[8]}) \rangle$ are the corresponding color-singlet and color-octet LDMEs of the mesons. In what follows, we disregard the contributions proportional to the color-octet LDMEs, as they are formally suppressed by $\mathcal{O}(v^4)$ in NRQCD [3, 8, 9] and are numerically small [89]. Potentially, the color-octet contributions could be enhanced near $z \approx 0$ due to the $1/z$ pole in Eq. (32). However, the residue of this pole is proportional to $\ln(\mu_0^2/4m_c^2)$, and for any reasonable choice of the scale μ_0 near $2m_c$, the contribution of $D_1^{\mathcal{Q}/b}$ remains negligible[90].

We can use the same framework to evaluate the fragmentation function for double-quarkonium production. In what follows, we will use the light-cone gauge $n \cdot A = A^+ = 0$, so the contributions of the gauge links (26) vanish identically. To find the amplitude of the process $q \rightarrow \mathcal{Q}_1 \mathcal{Q}_2 q$, we must take into account all the diagrams shown schematically in Fig. 2. In the heavy-quark mass limit, the quarks carry exactly half of the quarkonium momentum (neglecting internal motion), so the virtualities of all intermediate partons are fixed by the external kinematics and are large ($\sim M_{12}^2$), which justifies the perturbative evaluation. For vector charmonia ($J/\psi, \psi(2S)$), the last two diagrams are mediated by color-octet LDMEs and are thus negligibly small. The evaluation is simplest for fragmentation from b quarks, which receives contributions only from the first diagram in Fig. 2. We perform the evaluation in the heavy-quark mass limit using conventional NRQCD projectors (see details and the final analytic result in Appendix A). For the fragmentation of the leading charm quark, the evaluation follows the same lines, but the number of diagrams increases significantly. A direct evaluation becomes challenging because the full fragmentation function requires squaring the corresponding amplitude, which involves a sum of pairwise products of the different diagrams shown in Fig. 2. For this reason, we perform the calculation in the helicity basis, assuming that for unpolarized quarkonium, the cross section is dominated by contributions that do not flip the helicity (spin) of the initiating quark (see details in Appendix B).

The final results of this evaluation are presented in Fig. 3. In the upper row, we show the dihadron fragmentation function of $J/\psi J/\psi$. Since our perturbative approach may be unreliable in the near-threshold region, we analyze the contributions of different kinematic domains in the upper-left plot by introducing a lower cutoff on the invariant mass

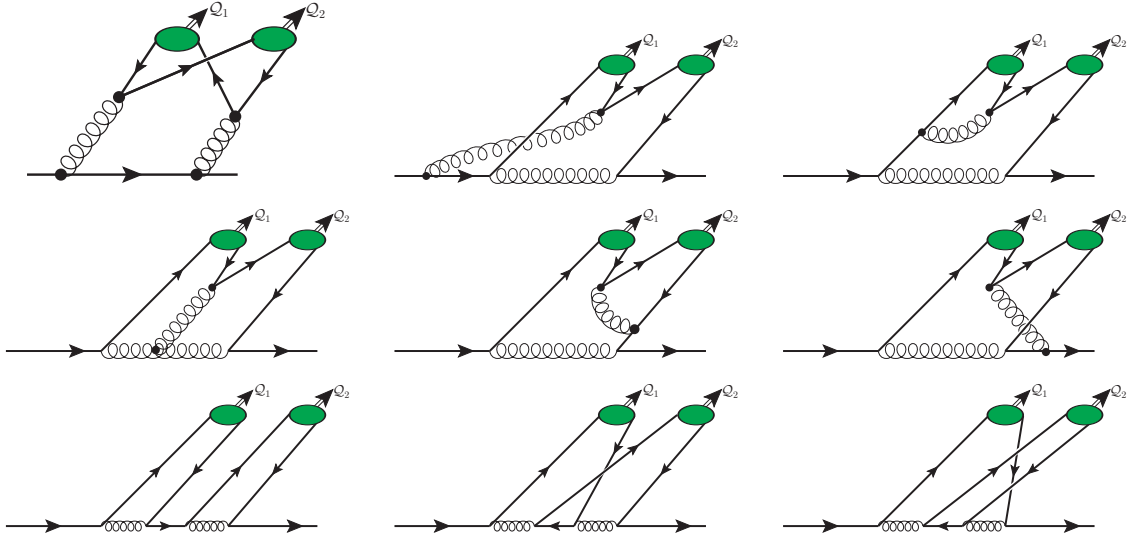


Figure 2. Leading-order diagrams contributing to the amplitude of the process $q \rightarrow Q_1 Q_2 q$ at leading order in α_s . Each diagram must be supplemented by a corresponding diagram with permuted mesons $Q_1 \leftrightarrow Q_2$. If the flavor of the initiating heavy quark does not match the constituent flavor of the quarkonia (e.g., $b \rightarrow J/\psi J/\psi b$), only the first diagram contributes. The last two diagrams contribute to J/ψ and $\psi(2S)$ production only via color-octet LDMEs. The full fragmentation function is proportional to a square of the sum of the corresponding amplitudes, and may be reduced to a sum of the pairwise products of different diagrams in the list.

M_{12} . While a fragmentation function with a kinematic cutoff lacks a strict formal definition and violates the principle of universality, it is useful for understanding the contributions from different regions and establishing a lower bound on the fragmentation function which comes from the perturbative domain. We observe that the cutoff suppresses the dihadron fragmentation function in the central region ($\zeta \approx 0$) but has almost no effect at large $|\zeta|$. Indeed, Eq. (28) shows that for such asymmetric configurations, the invariant mass is large even for $R_\perp = 0$, so the cutoff does not alter the result. The upper-right plot demonstrates that the dependence on the mass m of the initiating quark is very mild. This is not surprising, since in most expressions m appears in combination with parametrically larger variables (of the order of the invariant mass M_{12}^2). In the lower-left panel, a comparison of the red and blue surfaces shows that unequal quarkonium masses introduce a slight asymmetry in the ζ dependence of the dihadron fragmentation function. Finally, in the lower-right panel, we compare the dihadron fragmentation function with the product of two single-fragmentation functions. The latter is more than an order of magnitude smaller because single fragmentation from a bottom quark is governed by the small color-octet LDME, whereas single fragmentation from a charm quark requires the co-production of an additional undetected c or $\bar{c}$ quark for each fragmentation function [36], leading to strong phase-space suppression.

C. Quarkonia production via the fragmentation mechanism in the CGC

In this section, we show in detail that fragmentation in the CGC framework corresponds to a specific class of diagrams, shown schematically in Fig. 4. The main goal of this section is to demonstrate that the off-shellness of the heavy quarks does not prevent the representation of the cross section as a convolution of the dipole cross section (20) with the fragmentation functions (25) and (29). Furthermore, in the heavy-quark mass limit, there is no need to introduce an artificial lower cutoff on the transverse momenta of the quarks. However, this analysis does not apply to light quarks, since a large invariant mass of the quarkonium pair alone is insufficient to justify a perturbative treatment. For this reason, we consider only the contributions from heavy flavors (c and b), assuming that such events can be experimentally separated from the light-quark background.

For simplicity, in what follows we consider only the unpolarized (spin-averaged) production cross sections. Technically, this factorization in the CGC picture is straightforward to justify and closely follows the derivation of the $Q\bar{Q}$ cross section in Refs. [66, 70, 71]. For definiteness, we demonstrate how factorization occurs for one specific diagram which corresponds to dihadron fragmentation, assuming that the proof can be extended to all other contributions. For convenience, we perform the analysis in the reference frame where the incoming gluon q is on-shell and moves in the plus direction, $q^\mu = (q^+, 0, \mathbf{0}_\perp)$, assuming that all momenta in Eqs. (4)–(6) can be transformed to this frame by

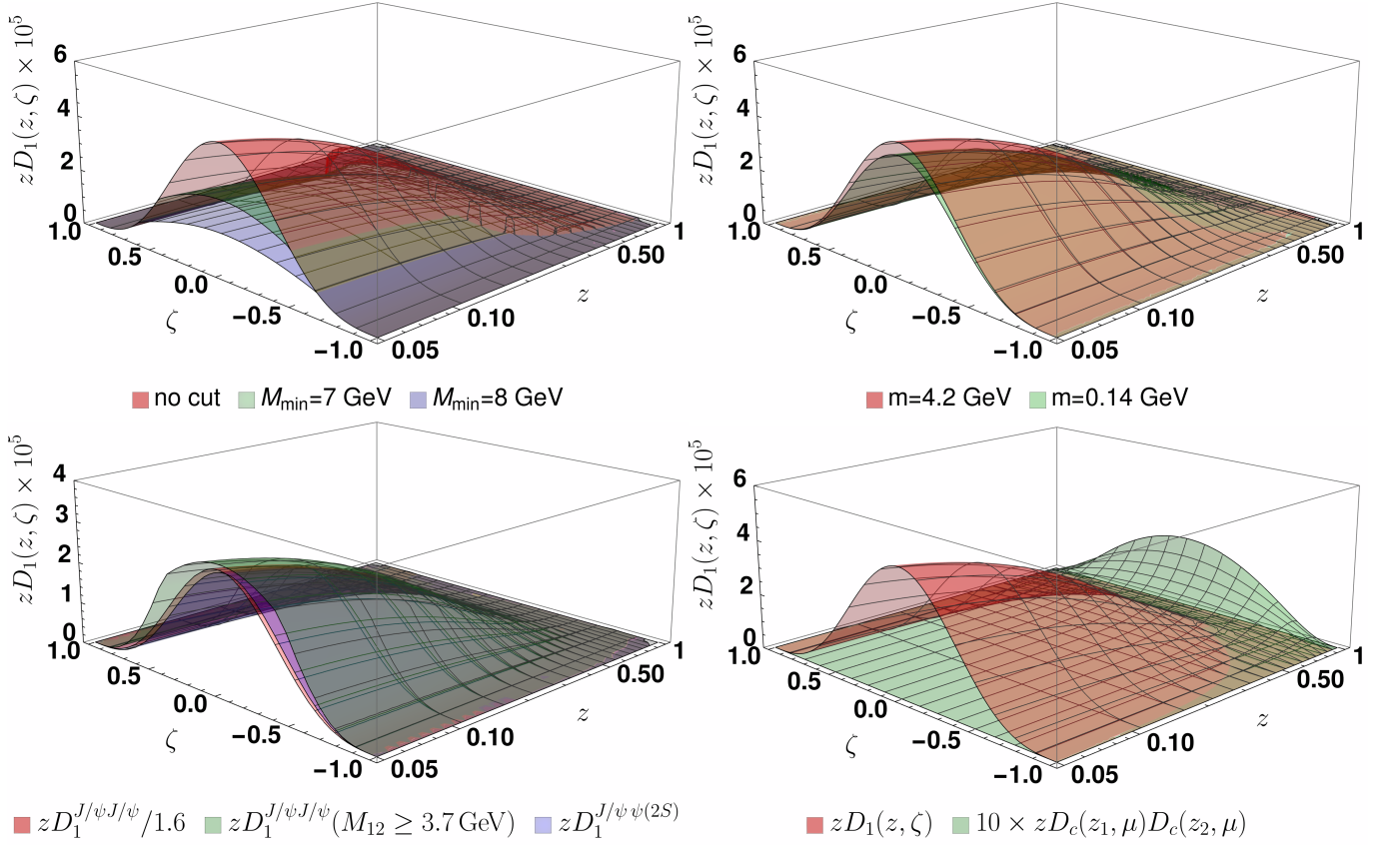


Figure 3. (Color online) Dihadron fragmentation function $zD_1(z, \zeta)$ as a function of the variables (z, ζ) . Upper row: Dihadron fragmentation function for $J/\psi J/\psi$ for different choices of the invariant mass cutoff $M_{\min}$ and the initiating quark mass m . Lower-left plot: Comparison of the dihadron fragmentation functions for $J/\psi J/\psi$ and $J/\psi \psi(2S)$. For the $J/\psi J/\psi$ case, we show results with an invariant mass cutoff (green) and with an arbitrary normalization factor $N \approx 1/1.6$ to match the normalization (red). Lower-right plot: Comparison of the dihadron fragmentation function for $J/\psi J/\psi$ with the product of two single-fragmentation functions $c \rightarrow J/\psi$.

a transverse boost $\mathbf{k}_{a\perp} \rightarrow \mathbf{k}_{a\perp} - \mathfrak{z}_a \mathbf{q}_\perp$, where $\mathfrak{z}_a = k_a^+/q^+$ is the light-cone momentum fraction of the corresponding particle (which is invariant under such boosts). The virtual b quark subsequently fragments in the vacuum into a final-state b quark with momentum k_1 and two heavy charmonia with momenta p_1 and p_2 . Momentum conservation allows us to reconstruct the momentum of the quark prior to fragmentation (immediately after its interaction with the shockwave):

$$K = k_1 + p_1 + p_2. \quad (33)$$

In the color-singlet NRQCD framework, we neglect the internal relative motion of the constituent quarks, so the momentum of each charmonium is shared equally between its constituent c and $\bar{c}$ quarks. The momenta of the gluons connecting the charm and bottom lines can then be determined from momentum conservation as

$$q_1 = q_2 = \frac{p_1 + p_2}{2}, \quad q_1^2 = q_2^2 = \frac{M_{12}^2}{4}. \quad (34)$$

Using the CGC Feynman rules, the cross-section of the quarkonium pair production can be represented as

$$\frac{d\sigma(g + p \rightarrow Q_1 + Q_2 + X)}{dy_1 dy_2} = \frac{1}{16\pi} \int_0^1 d\alpha \int \prod_{j=1}^2 \frac{d^2 \mathbf{p}_j}{2(2\pi)^3} \frac{d^2 \mathbf{k}_j}{2(2\pi)^3} \langle |\mathcal{A}^a|^2 \rangle_Y, \quad (35)$$

where $\alpha = K^+/q^+$, and the variables k_1 and k_2 are the momenta of active quark (after fragmentation) and the spectator antiquark, respectively. The bracket $\langle \dots \rangle_Y$ stands for the averaging over the color sources discussed in

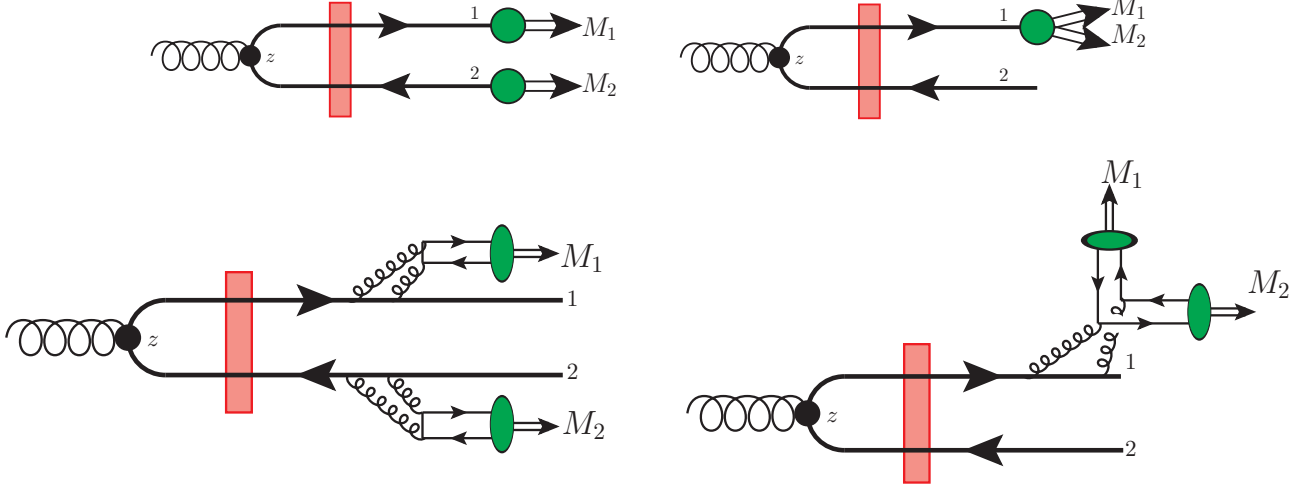


Figure 4. Upper row: Representative diagrams illustrating the single- and double-fragmentation contributions to inclusive heavy-quark pair hadroproduction in the CGC framework at leading order in α_s (left and right panels, respectively). Lower row: Representative diagrams corresponding to single- and dihadron fragmentation. For the quarkonium states considered here, both the quark and antiquark produced after the shockwave contribute equally to the fragmentation. The red block represents the shockwave, which can interact either with the quarks or the incoming gluon.

Section II A. The notation $\mathcal{A}^a$ is used for the amplitude of the $g \rightarrow Q_1 Q_2 b \bar{b}$ subprocess in the field of the shockwave. The latter amplitude can be represented as

$$\mathcal{A}^a = \sqrt{4\pi\alpha_s} \int \frac{d^4 k}{(2\pi)^4} \int d^2 \mathbf{x}_1 d^2 \mathbf{x}_2 e^{i(\mathbf{k}_\perp - \mathbf{K}_\perp) \cdot \mathbf{x}_1 - i(\mathbf{k}_{2\perp} - \mathbf{k}_\perp) \cdot \mathbf{x}_2} [U(\mathbf{x}_1) t^a U^\dagger(\mathbf{x}_2) - \mathcal{U}_{ab}(\mathbf{b}_{12}) t^b]_{\bar{c}c'} \times \quad (36)$$

$$\times (2\pi) \delta(k^+ - K^+) \bar{u}(k_1) \mathcal{M}_{\text{frag}}^{c,c'}(k_1, p_1, p_2; K) S_F(K) \gamma^+ S_F(k) \not{\epsilon}(q) S_F(q - k) \gamma^+ v(k_2)$$

where $\varepsilon_\mu^{(\lambda)}(q)$ is the polarization vector of the incoming (primordial) gluon with polarization λ , and $\not{\epsilon} \equiv \varepsilon_\mu^{(\lambda)} \gamma^\mu$. The contribution $\mathcal{U}_{ab}(\mathbf{b}_{12})$ corresponds to the interactions of the incoming gluon with transverse coordinate $\mathbf{b}_{12}$ defined in (16). The notation $\mathcal{M}_{\text{frag}}^{c,c'}(k_1, p_1, p_2; K)$ stands for the (cut) amplitude of the $b^{(c')} \rightarrow b^{(c)} Q_1 Q_2$ process (the sum of all the appropriate contributions shown in the Figure 3 is implied), c' and c are color indices of quark before and after fragmentation [91], and

$$S_F(\ell) = i \frac{\not{\ell} + m}{\ell^2 - m^2}, \quad (37)$$

is the free Dirac propagator. It is convenient to rewrite the quark propagators connected to the shockwave (which are multiplied by γ^+) as

$$S(\ell) = \frac{i(\not{\ell} + m)}{\ell^2 - m^2 + i0} = \theta(\ell^+) \frac{i(\not{\ell}'_Q + m)}{\ell^2 - m^2 + i0} - \theta(-\ell^+) \frac{i(\not{\ell}'_{\bar{Q}} - m)}{\ell^2 - m^2 + i0} + \frac{\left(\ell^- - \frac{\ell_\perp^2 + m^2}{2\ell^+}\right) \gamma^+}{\ell^2 - m^2 + i0} = \quad (38)$$

$$= \frac{\theta(\ell^+) \sum_h u_h(\ell'_Q) \bar{u}_h(\ell'_Q) - \theta(-\ell^+) \sum_h v_h(\ell'_{\bar{Q}}) \bar{v}_h(\ell'_{\bar{Q}})}{\ell^2 - m^2 + i0} + \frac{\gamma^+}{2\ell^+},$$

where the prime ($'$) denotes that the minus-component of the vector is adjusted to satisfy the on-shell condition $\ell^2 = m^2$, namely

$$\ell' = \left(\ell^+, \frac{\ell_\perp^2 + m^2}{2\ell^+}, \ell_\perp \right), \quad \ell_Q \equiv \ell', \quad \ell'_{\bar{Q}} = -\ell'. \quad (39)$$

The last term in Eq. (38) corresponds to the instantaneous propagator in light-cone perturbation theory [19]. Since the shockwave interaction vertices (11)–(12) contain the matrix γ^+ , the identity $\gamma^+ \gamma^+ = 0$ allows us to neglect the

instantaneous term in the propagators connected to the shockwave. The product of the quark propagators connected to the shockwave can then be simplified using the identities [19]

$$\bar{u}_{h_1}(\ell_1) \gamma^+ u_{h_2}(\ell_2) = \bar{v}_{h_1}(\ell_1) \gamma^+ v_{h_2}(\ell_2) = 2\sqrt{\ell_1^+ \ell_2^+} \delta_{h_1, h_2}, \quad (40)$$

$$S(\ell_1) \gamma^+ S(\ell_2) = 2\sqrt{\ell_1^+ \ell_2^+} \frac{\theta(\ell_1^+) \sum_h u_h(\ell_1') \bar{u}_h(\ell_2') - \theta(-\ell_1^+) \sum_h v_h(-\ell_1') \bar{v}_h(-\ell_2')}{(\ell_1^2 - m^2 + i0)(\ell_2^2 - m^2 + i0)}. \quad (41)$$

The dependence on k^- in the integrand of Eq. (36) is isolated in the product $S_F(k) \not{\epsilon}(q) S_F(q-k)$. Integration over this variable is straightforward using the Cauchy's residue theorem:

$$\begin{aligned} & \int_{-\infty}^{+\infty} \frac{dk^-}{2\pi} \frac{1}{(2k^+ k^- - \mathbf{k}_\perp^2 - m^2 + i0) \left(2(q^+ - k^+) (q^- - k^-) - (\mathbf{q}_\perp - \mathbf{k}_\perp)^2 - m^2 + i0 \right)} = \\ & = -\frac{\Theta(q^+ - k^+) \Theta(k^+)}{4k^+ (q^+ - k^+)} \frac{i}{q^- - \frac{\mathbf{k}_\perp^2 + m^2}{2k^+} - \frac{(\mathbf{q}_\perp - \mathbf{k}_\perp)^2 + m^2}{2(q^+ - k^+)}}. \end{aligned} \quad (42)$$

The Heaviside step functions $\Theta(\dots)$ in Eq. (42) restrict the range of the k^+ integration and arise from the requirement that the integrand has poles on both sides of the real k^- axis to yield a non-zero result. Integrating over $\mathbf{k}_\perp$ subsequently yields the Macdonald functions K_0 and K_1 , so that the amplitude (36) reduces to

$$\begin{aligned} \mathcal{A}^a &= \bar{u}_h(k_1) \mathcal{M}_{\text{frag}}^{c, c'} S_F(K) \gamma^+ u_{h'}(K) \times \\ & \times \int d^2 \mathbf{x}_1 d^2 \mathbf{x}_2 e^{-i\mathbf{K}_\perp \cdot \mathbf{x}_1 - i\mathbf{k}_{2\perp} \cdot \mathbf{x}_2} \Psi_{g \rightarrow \bar{b}b}^{(\lambda, h', \bar{h})}(\alpha, \mathbf{x}_1 - \mathbf{x}_2) [U(\mathbf{x}_1) t^a U^\dagger(\mathbf{x}_2) - \mathcal{U}_{ab}(\mathbf{b}_{12}) t^b]_{\bar{c}, c'}, \end{aligned} \quad (43)$$

where λ is the helicity of the incoming gluon, $\bar{c}, \bar{h}$ are the color and helicity indices of the produced antiquark, and h', h are the helicities of the quark before and after the fragmentation. The gluon splitting wave function $\Psi_{g \rightarrow \bar{b}b}^{(\lambda, h, \bar{h})}$ at leading order in α_s is given by [73, 78]:

$$\begin{aligned} \Psi_{g \rightarrow \bar{b}b}^{(\lambda, h, \bar{h})}(\alpha, r) &= -\frac{i\sqrt{4\pi\alpha_s}}{\sqrt{2k^+ 2(q^+ - k^+)}} \int \frac{d^2 \mathbf{k}_\perp}{(2\pi)^2} e^{i\mathbf{k}_\perp \cdot \mathbf{r}} \frac{\bar{u}_h(k^+, \mathbf{k}_\perp) \not{\epsilon}_\lambda(q) v_{\bar{h}}(q^+ - k^+, -\mathbf{k}_\perp)}{q^- - \frac{q^+(\mathbf{k}_\perp^2 + m^2)}{2k^+(q^+ - k^+)}} \Bigg|_{q^-=0, \quad k^+=\alpha q^+} \\ &= \frac{\sqrt{8\pi\alpha_s}}{2\pi} [-i\lambda e^{-i\lambda\phi_r} (\alpha\delta_{h,\lambda}\delta_{\bar{h},-\lambda} - (1-\alpha)\delta_{h,-\lambda}\delta_{\bar{h},\lambda}) mK_1(mr) + m\delta_{h,\lambda}\delta_{\bar{h},\lambda} K_0(mr)]. \end{aligned} \quad (44)$$

To obtain the complete scattering amplitude, we must add the contributions where the shockwave interacts directly with the incoming gluon, as well as the terms where the charmonia are produced from the antiquark. This procedure yields the total amplitude:

$$\begin{aligned} \mathcal{A}_{\text{tot}}^a &= \frac{1}{2\sqrt{\alpha(1-\alpha)}} \int d^2 \mathbf{x}_1 d^2 \mathbf{x}_2 \Psi_{g \rightarrow \bar{b}b}^{(\lambda, h', \bar{h}')}(\alpha, \mathbf{x}_1 - \mathbf{x}_2) [U(\mathbf{x}_1) t^a U^\dagger(\mathbf{x}_2) - \mathcal{U}_{ab}(\mathbf{b}_{12}) t^b]_{\bar{c}', c'} \times \\ & \times \left[\delta_{\bar{h}, \bar{h}'} \delta_{\bar{c}, \bar{c}'} \bar{u}_{h'}(k_1) \mathcal{M}_{\text{frag}}^{c', c}(k_1, p_1, p_2; K) S_F(K) \gamma^+ u_h(K') e^{-i\mathbf{K}_\perp \cdot \mathbf{x}_1 - i\mathbf{k}_{2\perp} \cdot \mathbf{x}_2} + \right. \\ & \left. + \delta_{h, h'} \delta_{c, c'} \bar{v}_{\bar{h}}(K') \gamma^+ S_F(K) \mathcal{M}_{\text{frag}}^{\bar{c}, \bar{c}'}(-k_2, p_1, p_2; -K) v_{\bar{h}'}(k_2) e^{-i\mathbf{K}_\perp \cdot \mathbf{x}_2 - i\mathbf{k}_{1\perp} \cdot \mathbf{x}_1} \right]. \end{aligned} \quad (45)$$

To obtain the cross section, we square the amplitude (45), sum over final-state color and spin indices, and average over the initial-state spin and color configurations. In general, this procedure leads to a convolution of the spin density matrices of the produced $Q\bar{Q}$ system with the helicity-dependent matrix $\hat{\rho}_{\mathbf{c}, \mathbf{c}'} = \sum_{\mathbf{j}} \hat{\mathcal{M}}_{\text{frag}}^{\mathbf{j}\mathbf{c}} \hat{\mathcal{M}}_{\text{frag}}^{*\mathbf{j}\mathbf{c}'}$, where we introduced the joint index notation $\mathbf{c} = (c, h)$, $\mathbf{c}' = (c', h')$ for color and helicity indices, and defined a shorthand notation [92]

$$\hat{\mathcal{M}}_{\text{frag}}^{\mathbf{c}, \mathbf{c}'}(k_1, p_1, p_2, K) = \bar{u}_h(k_1) \mathcal{M}_{\text{frag}}^{c, c'}(k_1, p_1, p_2; K) S_F(K) \gamma^+ u_{h'}(K'). \quad (46)$$

In what follows, we focus on the dominant unpolarized, color-singlet contributions, where the main contribution comes from $\mathbf{i} = \mathbf{i}'$. We observe that the amplitude $\mathcal{M}_{\text{frag}}^{\mathbf{j}\mathbf{i}}$ relates directly to the definition of the dihadron fragmentation function in Eq. (29) via

$$\int \frac{d^2 \mathbf{K}_\perp}{2(2\pi)^3} \frac{d^2 \mathbf{R}_\perp}{2(2\pi)^3} \sum_{\mathbf{i}, \mathbf{j}} \left| \hat{\mathcal{M}}_{\text{frag}}^{\mathbf{j}\mathbf{i}}(k_1, p_1, p_2; K) \right|^2 = \frac{1-z}{z} D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta), \quad (47)$$

where the integration variables $\mathbf{K}_\perp$ and $\mathbf{R}_\perp$ represent the transverse momentum of the leading quark and the relative transverse momentum of the individual quarkonium states with respect to the pair momentum $P = p_1 + p_2$, respectively. The prefactor on the right-hand side of Eq. (47) arises from the different phase-space factors for the production of a heavy quarkonium pair versus a bare quark pair[93]. The lower cutoff on the invariant mass M_{12} discussed in Eq. (9) translates into a restriction on the phase space:

$$R_\perp^2 \geq \frac{M_{\min}^2 (1 - \zeta^2) - 2(1 - \zeta)M_1^2 - 2(1 + \zeta)M_2^2}{4}. \quad (48)$$

The cross section for the dihadron fragmentation mechanism can then be expressed as

$$\frac{d\sigma(g + p \rightarrow \mathcal{Q}_1 + \mathcal{Q}_2 + X)}{dy_1 dy_2} \approx \int_{z_{\min}}^1 \frac{dz}{z^2} \left[D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta) + D_1^{\mathcal{Q}_1 \mathcal{Q}_2/\bar{q}}(z, \zeta) \right] \frac{d\sigma(g + p \rightarrow Q + \bar{Q} + X)}{d\tilde{y}_1 d\tilde{y}_2}, \quad (49)$$

where the tilde in the right-hand side indicates that the rapidities of the heavy quarks must be evaluated with respect to the parent quark, being effectively shifted as $\tilde{y}_i = y_i + \ln(1/z)$, while the parameter ζ is defined as $\zeta = \tanh(y_1 - y_2)$. The lower bound $z_{\min}$ in the integral of Eq. (49) is determined by the requirement that the light-cone momentum of the parent quark does not exceed the momentum of the incoming gluon. The light-cone momentum fractions x_1 and x_2 carried by the gluon from the projectile and exchanged with the target, respectively, are given by

$$x_1^{(\text{DF})} = \frac{q^+}{P_1^+} = \frac{M_1^\perp e^{y_1}}{\alpha z_1 \sqrt{s}} = \frac{M_2^\perp e^{y_2}}{\alpha z_2 \sqrt{s}}, \quad (50)$$

$$x_2^{(\text{DF})} = \frac{\sum p_{\text{out}}^- - \sum p_{\text{in}}^-}{P_2^-} \approx \frac{\bar{z} M_{12}^2 + z m^2 (1 - \alpha z)}{z (1 - z) \alpha (1 - \alpha) x_1 s}, \quad (51)$$

where we neglected the transverse-momentum dependence in the dipole amplitude due to its relatively mild effect on x_2 and mild dependence of the dipole cross-section on x_2 . In view of the C -parity we may expect that the contributions of the quark and antiquark in (49) will coincide.

Similarly, we can calculate the cross section for quarkonium pair production via the single-fragmentation mechanism. This process occurs independently for each of the produced quarkonia and is described by single-fragmentation functions $D_1^{\mathcal{Q}_i/q}(z_i)$, which represent the probability for a heavy quark with momentum fraction z_i to form a quarkonium state $\mathcal{Q}_i$. The corresponding cross section is given by

$$\frac{d\sigma(g + p \rightarrow \mathcal{Q}_1 + \mathcal{Q}_2 + X)}{dy_1 dy_2} = \int_{z_{\min}}^1 \frac{dz_1}{z_1^2} \int_{z_{\min}}^1 \frac{dz_2}{z_2^2} D_1^{\mathcal{Q}_1/q}(z_1) D_1^{\mathcal{Q}_2/q}(z_2) \frac{d\sigma(g + p \rightarrow Q + \bar{Q} + X)}{d\bar{y}_1 d\bar{y}_2}, \quad (52)$$

where $\bar{y}_i = y_i + \ln(1/z_i)$ are the rapidities of the heavy quarks prior to fragmentation. The lower bounds on the z_1 and z_2 integrals are determined by the condition that the sum of the light-cone energies of the Q and $\bar{Q}$ quarks does not exceed the energy of the incoming gluon,

$$q^+ \geq \frac{1}{\sqrt{2}} \left(\frac{M_1^\perp e^{y_1}}{z_1} + \frac{M_2^\perp e^{y_2}}{z_2} \right). \quad (53)$$

The corresponding light-cone momentum fractions x_1 and x_2 are given by

$$x_1^{(\text{SF})} = \frac{q^+}{P_1^+} = \frac{M_1^\perp e^{y_1}}{\alpha z_1 \sqrt{s}} + \frac{M_2^\perp e^{y_2}}{(1 - \alpha) z_2 \sqrt{s}}, \quad (54)$$

$$x_2^{(\text{SF})} = \frac{\sum p_{\text{out}}^- - \sum p_{\text{in}}^-}{P_2^-} = \frac{\Delta q^-}{P^+} \approx \frac{m^2}{x_1 s \alpha (1 - \alpha)}. \quad (55)$$

Finally, we note that our analysis considers only the specific contributions that can be clearly attributed to either single- or dihadron fragmentation. The full cross section also includes various interference terms, such as the interference between single- and double-fragmentation amplitudes, or between dihadron fragmentation from different quarks (the second and the third lines in (45)). While these interference terms are suppressed when the quark and antiquark are well separated in rapidity or transverse momentum, they can double the overall contribution of the fragmentation mechanism in some regions. Since they do not reduce to a standard fragmentation convolution, we will not discuss them in what follows.

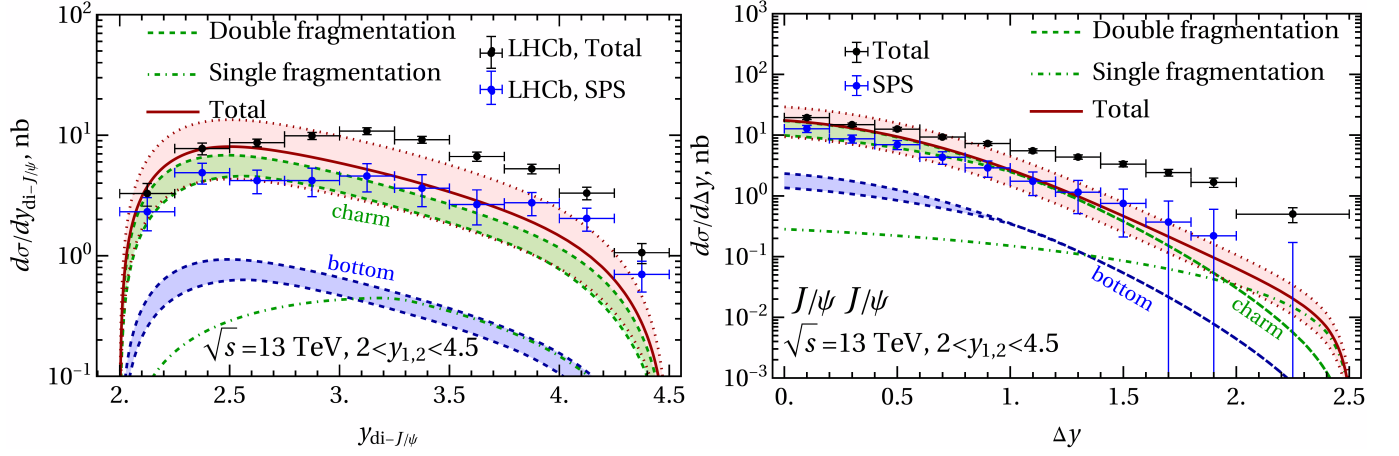


Figure 5. (Color online) Contributions of the single- and double-fragmentation mechanisms to inclusive $J/\psi J/\psi$ production from heavy quarks. For double fragmentation, the upper and lower boundaries of the bands correspond to calculations using the full dihadron fragmentation function and a version with an additional invariant mass cutoff $M_{12}^{\min} = 2M_{J/\psi} + \Lambda \approx 7$ GeV, respectively, so that the band width reflects the near-threshold contribution. The green and blue bands represent the contributions of charm and bottom DiFFs, respectively. The solid curve labeled “Total” includes both single- and double-fragmentation contributions from charm and bottom quarks, with the colored band showing the uncertainty from varying the scale μ by a factor of two and varying the collinear gluon PDF parameterizations. Experimental data are from Ref. [81].

III. NUMERICAL ESTIMATES FOR THE CROSS SECTION

We focus on the production of S -wave charmonium pairs (J/ψ and $\psi(2S)$ mesons), which have very clear experimental signatures and whose color-singlet long-distance matrix elements (LDMEs) are known with reasonable precision [11, 22, 79]. As explained above, for the p_T -integrated cross sections, quadrupole amplitudes do not contribute, and all nuclear interactions are described by color-dipole scattering amplitudes. For definiteness, we employ the bCGC parameterization of the color-dipole amplitude with fit parameters from Ref. [80], corresponding to a charm quark mass of $m_c \approx 1.4$ GeV. For the forward collinear PDFs, we use the “HERAPDF20_LO_EIG” dataset evaluated at the scale $\mu = 2M_{J/\psi}$ (the same scale is used for α_s). Since our calculations are performed at leading order, we do not consider the explicit DGLAP evolution of the fragmentation functions, assuming that the LO expressions derived in Sec. IIB are evaluated at the same scale μ . We estimate the theoretical uncertainty due to choice of the scale varying it by a factor of two.

In Figs. 5–6, we compare the predicted contributions of the fragmentation mechanisms with experimental data from Refs. [81, 82]. We verified that in the relevant kinematics, the light-cone fractions satisfy $x_1 \geq 10^{-1}$ and $x_2 \leq 10^{-2}$, justifying the use of the dilute-dense approximation. In all plots the dot-dashed lines correspond to the independent (uncorrelated) single fragmentation of each quark into a quarkonium state, which remains negligible over the entire kinematic range. This suppression occurs because single fragmentation either proceeds via small color-octet LDMEs (for b quarks) or requires the co-production of two additional charmed quarks.

Since the perturbative treatment may lose reliability in the near-threshold region, we provide predictions using both the full fragmentation function and a version containing a cut on the invariant mass M_{12} , as discussed below Eq. (8). The suppression of heavy-quark contributions with increasing heavy-quark mass m approximately follows a power law $\sim 1/m^{2\gamma}$, and is explained by the suppression of the heavy-quark production cross section due to color transparency (as we discussed earlier, the dihadron fragmentation itself depends very weakly on the mass of the parent quark). The suppression of the cross section at large rapidity separation Δy arises from the suppression of the dihadron fragmentation function at large invariant masses M_{12} . The dependence on the rapidity y is determined by the interplay between the x_1 dependence of the collinear PDF, the x_2 dependence of the dipole amplitude, and the requirement that both charmonia fall within the acceptance of the LHCb detector. Because the cross section decreases rapidly with p_T , the fiducial cross section measured with an upper cutoff $p_T < 14$ GeV by the LHCb Collaboration practically coincides with the p_T -integrated cross section studied in our framework. We do not consider contributions from fragmentation of light quarks or gluons, because there is no hard scale to justify a perturbative treatment of the corresponding splitting subprocesses, such as $g \rightarrow q\bar{q}$ and $gg \rightarrow g$. Taken together, the charm and bottom contributions are sizable and qualitatively reproduce the shape of the observed rapidity dependence. In Fig. 7, we also present predictions for $\psi(2S)\psi(2S)$ production, which can be tested in future experimental studies.

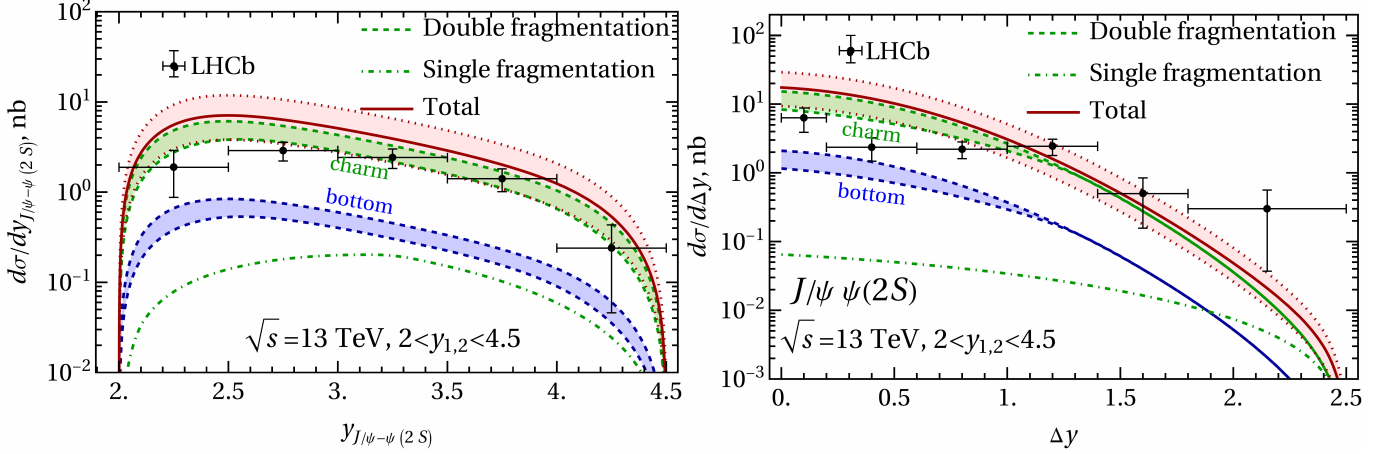


Figure 6. (Color online) Contributions of the single- and double-fragmentation mechanisms to inclusive $J/\psi \psi(2S)$ production from heavy c and b quarks. For double fragmentation, the upper and lower boundaries of the bands correspond to calculations with the full dihadron fragmentation function and with an additional invariant mass cutoff $M_{12}^{\min} = M_{J/\psi} + M_{\psi(2S)} + \Lambda \approx 7.8$ GeV, respectively. The green and blue bands correspond to the contributions of charm and bottom DiFFs, respectively. The solid curve labeled “Total” shows the combined single- and double-fragmentation contributions from charm and bottom quarks, with the band representing the uncertainty from varying the scale μ by a factor of two and varying the collinear gluon PDF parameterizations. Experimental data are from Ref. [82].

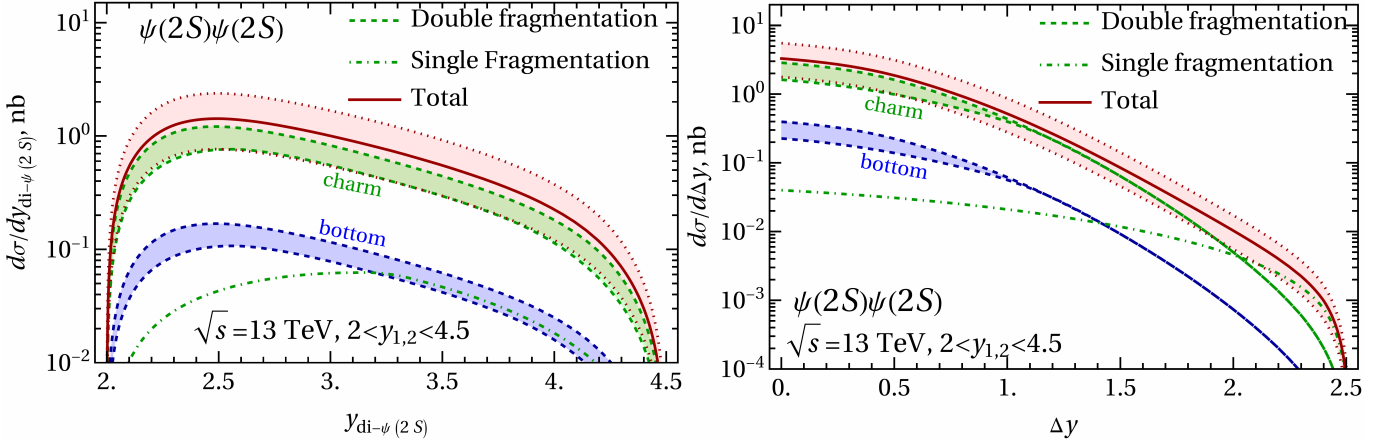


Figure 7. (Color online) Contributions of the single- and double-fragmentation mechanisms to inclusive $\psi(2S) \psi(2S)$ production from heavy c and b quarks. For double fragmentation, the upper and lower boundaries of the bands represent the calculations with the full dihadron fragmentation function and with an additional invariant mass cutoff $M_{12}^{\min} = 2M_{\psi(2S)} + \Lambda \approx 8.5$ GeV, respectively. The green and blue bands represent the contributions of charm and bottom DiFFs, respectively. The solid curve labeled “Total” represents the combined single- and double-fragmentation contributions from charm and bottom quarks, with the band showing the uncertainty from the scale μ variation and the collinear gluon PDF parameterizations.

IV. CONCLUSIONS

In this manuscript, we evaluated the dihadron fragmentation function and analyzed the contribution of fragmentation mechanisms to the inclusive hadroproduction of heavy quarkonium pairs. We focused on the p_T -integrated cross sections, which are controlled by forward dipole amplitudes. Our results indicate that while single-quark fragmentation represents only a small correction, dihadron fragmentation provides a significant contribution that is on par with other production mechanisms. These findings remain robust under variations of the renormalization scale, parameterizations of the collinear gluon PDF, and the forward dipole scattering amplitude, as well as under conservative lower cuts on the invariant mass M_{12} of the produced system.

Since at present our results indicate that the dihadron fragmentation mechanism gives a significant contribution, the

sum of this mechanism together with direct SPS charmonia pair production (via $g + \text{shockwave} \rightarrow (c\bar{c})(c\bar{c}) \rightarrow \mathcal{Q}_1\mathcal{Q}_2$ subprocess), will exceed the experimental data, thus not leaving space for the contribution of the Double Parton Scattering, and won't allow extraction of the effective cross-section σ_{eff} that encodes its magnitude. This tension underscores the importance of next-to-leading-order (NLO) corrections. Indeed, it is well known that NLO corrections can suppress the $Q\bar{Q}$ production cross section by up to a factor of two [70, 71, 83]. A similar suppression in the dihadron fragmentation channel would restore the space for DPS contributions, allowing for a meaningful extraction of σ_{eff} . However, a systematic evaluation of NLO corrections to the fragmentation mechanism requires computing the corrections to both the single- and dihadron fragmentation functions, as well as accounting for real emission, which is beyond the scope of this work. Furthermore, such an analysis would require modeling multipole (quadrupole and sextupole) operators, which are currently poorly constrained phenomenologically.

Although we have focused on fragmentation in pp collisions, extending this framework to pA collisions is straightforward. Since the fragmentation time t_f exceeds the interaction time t_h by a factor of $1/x_2 \sim 10^2$ in the considered kinematics, nuclear medium effects should have only a minor influence on the fragmentation functions, primarily affecting the forward dipole scattering amplitude (19).

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Appendix A: Evaluation of the diquarkonium fragmentation function

In this appendix, we provide the technical details of the evaluation of the dihadron fragmentation function. We perform the evaluations in the heavy-quark mass limit and employ conventional NRQCD projectors. We focus on b quarks, for which the evaluation is the simplest. Since the momenta of the quarkonia are shared equally between the constituent charm quarks in the diagrams, both gluons carry the same momentum $\ell = (P_1 + P_2)/2$ and are significantly off-shell, namely $\ell^2 = M_{12}^2/4$. The evaluation of the corresponding diagram is straightforward and yields

$$D_1^{\mathcal{Q}_1\mathcal{Q}_2/q}(z, \zeta) = C_F^2 \int \frac{d^2\mathbf{K}_\perp}{2k'^+(2\pi)^3} \frac{d^2\mathbf{R}_\perp}{(2\pi)^2} V_{\mu\nu} V_{\alpha\beta}^* D_{\mu\mu'}(\ell) D_{\alpha\alpha'}(\ell) D_{\beta\beta'}(\ell) D_{\nu\nu'}(\ell) \times \text{Tr} \left(\gamma_+ S(K) \gamma^{\mu'} S(K - \ell/2) i\gamma^{\nu'} (\not{K} - \not{P}_1 - \not{P}_2 + m) \gamma^{\alpha'} S(K - \ell/2) \gamma^{\beta'} S(K) \right) \Big|_{K^+ = P_h^+/z}, \quad (\text{A1})$$

where the functions $V_{\mu\nu}$ correspond to the amplitude of the $gg \rightarrow \mathcal{Q}_1\mathcal{Q}_2$ subprocess (the upper fermion loop in the first diagram in Fig. 2), $D_{\mu\mu'}$ are the gluon propagators, and K^μ is the momentum of the parent quark, whose minus-component K^- is fixed by the requirement that the final-state quark is on-shell,

$$K^- = P_1^- + P_2^- + z \frac{\mathbf{k}_{1,\perp}^2 + m^2}{2P_h^+(1-z)}. \quad (\text{A2})$$

To evaluate the functions $V_{\mu\nu}$, we need to project the $Q\bar{Q}$ pairs onto states with definite color and spin. According to potential models and NRQCD, the dominant Fock states of the J/ψ and $\psi(2S)$ are the color-singlet $Q\bar{Q}$ pairs in the $^3S_1^{[1]}$ state. As discussed in Refs. [8, 9, 84], the projectors onto color-singlet and color-octet states are given respectively by

$$\left(\mathcal{P}^{[1]} \right)_{ij} = \frac{\delta_{ij}}{\sqrt{N_c}}, \quad \left(\mathcal{P}_b^{[8]} \right)_{ij} = \sqrt{2} (t^b)_{ij}, \quad b = 1, \dots, 8. \quad (\text{A3})$$

The projections onto a state with definite total spin S and its projection S_z can be found using the appropriate Clebsch-Gordan coefficients [8, 9, 84]:

$$\begin{aligned} \hat{\mathcal{P}}_{SS_z} &= \sum_{s_1, s_2} \left\langle \frac{1}{2} s_1 \frac{1}{2} s_2 \middle| 1 S_z \right\rangle v \left(\frac{P}{2} - \mathbf{q}, s_2 \right) \bar{u} \left(\frac{P}{2} + \mathbf{q}, s_1 \right) = \\ &= \frac{-1}{2\sqrt{2}} \left(\frac{\not{P}}{2} - \not{\mathbf{q}} - m_c \right) \not{\epsilon}_{\mathcal{Q}}(P) \left(\frac{\not{P}}{2} + \not{\mathbf{q}} + m_c \right), \end{aligned} \quad (\text{A4})$$

where P is the momentum of the produced quarkonium, $\mathbf{q} \approx 0$ is the momentum of relative motion of the quarks inside the quarkonium, and ε_Q is the polarization vector of the vector meson. Combining these projectors with the proper color-singlet LDMEs and neglecting the relative momentum $\mathbf{q}$, we obtain the effective heavy-quark projectors onto the J/ψ and $\psi(2S)$ states:

$$\left(\hat{\mathcal{P}}_Q^{[1]}\right)_{ij} \approx \sqrt{\frac{\langle \mathcal{O}_Q^{[1]} \rangle}{m_c}} \frac{\delta_{ij}}{4N_c} \not{P} \left(\frac{\not{P}}{2} + m_c\right) \quad (\text{A5})$$

where $\langle \mathcal{O}_Q^{[1]} \rangle \equiv \langle \mathcal{O}_Q^{[1]}(^3S_1^{[1]}) \rangle$ are the corresponding color-singlet long-distance matrix elements for J/ψ and $\psi(2S)$ mesons. Phenomenological estimates, for example, those based on the leptonic decay width of $J/\psi \rightarrow e^+e^-$, suggest that $\langle \mathcal{O}_{J/\psi}^{[1]}(^3S_1^{[1]}) \rangle \approx 1.16 \text{ GeV}^3$ and $\langle \mathcal{O}_{\psi(2S)}^{[1]}(^3S_1^{[1]}) \rangle \approx 0.76 \text{ GeV}^3$ [22, 85].

Using these projectors and taking the traces over Dirac and color indices, we obtain the functions $V_{\mu\nu}$:

$$\begin{aligned} V_{\mu\nu} = \text{Tr} \left[\hat{\mathcal{P}}_{Q_1}^{[1]} \gamma_\mu \hat{\mathcal{P}}_{Q_2}^{[1]} \gamma_\nu \right] &= \frac{\sqrt{\langle \mathcal{O}_{Q_1}^{[1]} \rangle \langle \mathcal{O}_{Q_2}^{[1]} \rangle}}{16 m_c N_c} \left[g^{\mu\nu} \left(4 (P_1 \cdot \varepsilon_2) (P_2 \cdot \varepsilon_1) + 2 \left((M_1 - M_2)^2 - M_{12}^2 \right) (\varepsilon_1 \cdot \varepsilon_2) \right) \right. \\ &\quad - 2 \left((M_1 - M_2)^2 - M_{12}^2 \right) (\varepsilon_1^\mu \varepsilon_2^\nu + \varepsilon_1^\nu \varepsilon_2^\mu) - 4 (\varepsilon_1^\mu P_2^\nu (P_1 \cdot \varepsilon_2) + \varepsilon_2^\mu P_1^\nu (P_2 \cdot \varepsilon_1)) \\ &\quad \left. + 4 (\varepsilon_1 \cdot \varepsilon_2) (P_2^\mu P_1^\nu + P_1^\mu P_2^\nu) - 4 (P_2^\mu (\varepsilon_1^\nu (P_1 \cdot \varepsilon_2)) + P_1^\mu (\varepsilon_2^\nu (P_2 \cdot \varepsilon_1))) \right] \end{aligned} \quad (\text{A6})$$

The evaluation of the remaining traces is straightforward but tedious, and was performed using FeynCalc version 10.2 [86, 87]. The summation over the polarizations of the final-state quarkonia was performed using the standard identity

$$\sum_a \varepsilon_\alpha^{(a)} \varepsilon_\beta^{(a)*} = g_{\alpha\beta} - \frac{P_\alpha P_\beta}{M^2} \quad (\text{A7})$$

The final result can be represented in a compact form as

$$D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta) = \mathcal{G} \left[\hat{c}_0 + \hat{c}_A \log(1 + \varrho) + \hat{c}_B \log\left(1 - \frac{z}{2} + \varrho\right) + \hat{c}_U \log\left(1 - \frac{z}{2}\right) \right], \quad (\text{A8})$$

where we introduced the shorthand notations

$$\varrho = \frac{m^2 z^2}{M_{\min}^2 (1 - z)}, \quad \mathcal{G} = \frac{2 C_F^2 \alpha_s^4 \langle \mathcal{O}_{Q_1}^{[1]} \rangle \langle \mathcal{O}_{Q_2}^{[1]} \rangle (1 - z) (1 - \zeta^2)}{15 N_c^2 m^8 m_c^2 M_{\min}^{12} z^{12}}, \quad (\text{A9})$$

and $M_{\min}^2 = 2((1 - \zeta)M_1^2 + (1 + \zeta)M_2^2)/(1 - \zeta^2)$ is the minimal value of the invariant mass M_{12} of the quarkonium pair at a given ζ . The coefficients $\hat{c}_0, \hat{c}_A, \hat{c}_B, \hat{c}_U$ are polynomial functions of the variables z and ζ , whose explicit forms are specified below. In the limit $z \rightarrow 0$, the function $D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta)$ increases only as $\mathcal{O}(1/z)$ due to algebraic cancellations of the leading poles. This behavior is explained by the increase of the phase space volume with the energy of the parent quark $q^+ = P_h^+/z$, suppressed by an additional factor of z in the definition (29).

On general grounds, we expect the fragmentation function $D_1^{\mathcal{Q}_1 \mathcal{Q}_2/q}(z, \zeta)$ should be invariant under the permutation of the two quarkonium states; specifically, it should remain unchanged under the simultaneous exchange of their masses $M_1 \leftrightarrow M_2$ and the swap of the kinematic variables $\zeta \rightarrow -\zeta$. To demonstrate that the resulting expression possesses the desired symmetry and to present the explicit expressions for the functions $\hat{c}_0, \hat{c}_A, \hat{c}_B, \hat{c}_U$ in a compact form, we introduce auxiliary variables that respect this symmetry. First, we define the constants

$$S = M_1^2 + M_2^2, \quad P = M_1 M_2, \quad D = M_1^2 - M_2^2, \quad (\text{A10})$$

that satisfy the identity $D^2 = S^2 - 4P^2$. Since D is odd under the permutation of the two mesons, it appears in the final result multiplied by odd powers of ζ , whereas the symmetric combinations S and P are multiplied by even powers of ζ .

The explicit expressions for the coefficients $\hat{c}_0, \hat{c}_A, \hat{c}_B, \hat{c}_U$ are given below:

$$\begin{aligned} \hat{c}_0 = & -384 m^2 M_{\min}^{10} Q_1 z^3 + 2304 m^2 M_{\min}^{10} Q_1 z^4 \\ & + z^5 \left(-96 m^2 M_{\min}^{10} (580 D \zeta P + 136 D \zeta S + 852 \zeta^2 P^2 - 580 P^2 - 213 \zeta^2 S^2 - 68 S^2) \right) \end{aligned} \quad (\text{A11})$$

$$\begin{aligned}
& -384m^4M_{\min}^8 \left(8P^2 + S^2 - 8D\zeta P - 2D\zeta S - 12\zeta^2P^2 + 3\zeta^2S^2 \right. \\
& \quad \left. - 2M_{\min}^2 (33D\zeta^3 - 9D\zeta - 24\zeta^2P + 16P - 21\zeta^2S + S) \right) \\
& + z^6 \left(144m^2M_{\min}^{10} (768D\zeta P + 154D\zeta S + 1076\zeta^2P^2 - 768P^2 - 269\zeta^2S^2 - 77S^2) \right. \\
& \quad \left. + 48m^4M_{\min}^8 (304P^2 + 38S^2 - 304D\zeta P - 76D\zeta S - 456\zeta^2P^2 + 114\zeta^2S^2 \right. \\
& \quad \left. + M_{\min}^2 (-2388D\zeta^3 + 644D\zeta + 1744\zeta^2P - 1136P + 1516\zeta^2S - 76S) \right) \\
& + z^7 \left(-120m^2M_{\min}^{10} (-1300P^2 - 98S^2 + 1300D\zeta P + 196D\zeta S + 1692\zeta^2P^2 - 423\zeta^2S^2) \right. \\
& \quad \left. - 16m^4M_{\min}^8 (2064P^2 + 231S^2 - 2064D\zeta P - 462D\zeta S - 2988\zeta^2P^2 + 747\zeta^2S^2 \right. \\
& \quad \left. + M_{\min}^2 (-14052D\zeta^3 + 3836D\zeta + 10876\zeta^2P - 6724P + 8614\zeta^2S - 474S) \right) \\
& \quad + 8m^6M_{\min}^6 (384P^2 + 48S^2 - 384D\zeta P - 96D\zeta S - 576\zeta^2P^2 + 144\zeta^2S^2 \\
& \quad - 96M_{\min}^2 (33D\zeta^3 - 9D\zeta - 24\zeta^2P + 16P - 21\zeta^2S + S) \\
& \quad + (5940\zeta^4 - 7560\zeta^2 + 2100) M_{\min}^4) \\
& + z^8 \left(12m^2M_{\min}^{10} (-13508P^2 - 640S^2 + 13508D\zeta P + 1280D\zeta S + 16068\zeta^2P^2 - 4017\zeta^2S^2) \right. \\
& \quad \left. + 8m^4M_{\min}^8 (5820P^2 + 498S^2 - 5820D\zeta P - 996D\zeta S - 7812\zeta^2P^2 + 1953\zeta^2S^2 \right. \\
& \quad \left. + M_{\min}^2 (-15740P - 1080S + 8950D\zeta + 27620\zeta^2P + 17660\zeta^2S - 31470D\zeta^3) \right) \\
& \quad - 4m^6M_{\min}^6 (2688P^2 + 336S^2 - 2688D\zeta P - 672D\zeta S - 4032\zeta^2P^2 + 1008\zeta^2S^2 \\
& \quad + M_{\min}^2 (-9792P - 672S + 5568D\zeta + 15168\zeta^2P + 13152\zeta^2S - 20736D\zeta^3) \\
& \quad + (37260\zeta^4 - 46200\zeta^2 + 12300) M_{\min}^4) \\
& + z^9 \left(-18m^2M_{\min}^{10} (-7004P^2 - 178S^2 + 7004D\zeta P + 356D\zeta S + 7716\zeta^2P^2 - 1929\zeta^2S^2) \right. \\
& \quad \left. - 8m^4M_{\min}^8 (5484P^2 + 285S^2 - 5484D\zeta P - 570D\zeta S - 6624\zeta^2P^2 + 1656\zeta^2S^2 \right. \\
& \quad \left. + M_{\min}^2 (-12080P - 675S + 6715D\zeta + 23600\zeta^2P + 10445\zeta^2S - 22245D\zeta^3) \right) \\
& \quad + 8m^6M_{\min}^6 (2224P^2 + 224S^2 - 2224D\zeta P - 448D\zeta S - 3120\zeta^2P^2 + 780\zeta^2S^2 \\
& \quad + M_{\min}^2 (-13650D\zeta^3 + 3850D\zeta + 11120\zeta^2P - 6800P + 8090\zeta^2S - 450S) \\
& \quad + (22245\zeta^4 - 25610\zeta^2 + 6005) M_{\min}^4) \\
& \quad + 2m^8M_{\min}^4 (21504P^2 + 2688S^2 - 21504D\zeta P - 5376D\zeta S - 32256\zeta^2P^2 + 8064\zeta^2S^2 \\
& \quad + M_{\min}^2 (-9216P - 3456S + 8064D\zeta + 36864\zeta^2P + 26496\zeta^2S - 44928D\zeta^3) \\
& \quad + (26640\zeta^4 - 12960\zeta^2 - 4080) M_{\min}^4) \\
& - 2m^2M_{\min}^4z^{10} \left(-48M_{\min}^6 (740D\zeta P + 26D\zeta S + 792\zeta^2P^2 - 740P^2 - 198\zeta^2S^2 - 13S^2) \right. \\
& \quad \left. + m^2M_{\min}^4 (-14316P^2 - 318S^2 + 14316D\zeta P + 636D\zeta S + 15588\zeta^2P^2 - 3897\zeta^2S^2 \right. \\
& \quad \left. + M_{\min}^2 (24244P + 834S - 12956D\zeta - 55756\zeta^2P - 14074\zeta^2S + 41952D\zeta^3) \right) \\
& \quad + 4m^4M_{\min}^2 (2200P^2 + 128S^2 - 2200D\zeta P - 256D\zeta S - 2712\zeta^2P^2 + 678\zeta^2S^2 \\
& \quad + M_{\min}^2 (-9237D\zeta^3 + 2921D\zeta + 9406\zeta^2P - 5314P + 4534\zeta^2S - 264S)
\end{aligned}$$

$$\begin{aligned}
& + (12660\zeta^4 - 12070\zeta^2 + 1830) M_{\min}^4 \Big) \\
& + m^6 \Big(48384P^2 + 6048S^2 - 48384D\zeta P - 12096D\zeta S - 72576\zeta^2 P^2 + 18144\zeta^2 S^2 \\
& \quad + M_{\min}^2 \Big(-14976P - 7776S + 15264D\zeta + 77184\zeta^2 P + 53856\zeta^2 S - 92448D\zeta^3 \Big) \\
& \quad + (54180\zeta^4 - 25320\zeta^2 - 7260) M_{\min}^4 \Big) \Big) \\
& + 2m^2 M_{\min}^2 z^{11} \Big(-54M_{\min}^8 (244D\zeta P + 12D\zeta S + 268\zeta^2 P^2 - 244P^2 - 67\zeta^2 S^2 - 6S^2) \\
& \quad + m^2 M_{\min}^6 \Big(-6660P^2 - 90S^2 + 6660D\zeta P + 180D\zeta S + 7020\zeta^2 P^2 - 1755\zeta^2 S^2 \\
& \quad + M_{\min}^2 (13992D\zeta^3 - 3676D\zeta - 22436\zeta^2 P + 7004P - 2774\zeta^2 S + 174S) \Big) \\
& \quad + m^8 \Big(15360P^2 + 1920S^2 - 15360D\zeta P - 3840D\zeta S - 23040\zeta^2 P^2 \\
& \quad + 5760\zeta^2 S^2 + 2304M_{\min}^2 (-3D\zeta^3 - D\zeta + 4\zeta^2 P + 4P + \zeta^2 S - S) \\
& \quad + (2160\zeta^4 + 1440\zeta^2 + 2160) M_{\min}^4 \Big) \\
& \quad + 8m^6 M_{\min}^2 \Big(4896P^2 + 504S^2 - 4896D\zeta P - 1008D\zeta S - 6912\zeta^2 P^2 \\
& \quad + 1728\zeta^2 S^2 + M_{\min}^2 (-7434D\zeta^3 + 762D\zeta + 6492\zeta^2 P - 228P + 4188\zeta^2 S - 648S) \\
& \quad + (4245\zeta^4 - 1730\zeta^2 - 355) M_{\min}^4 \Big) \\
& \quad + m^4 M_{\min}^4 \Big(5072P^2 + 88S^2 - 5072D\zeta P - 176D\zeta S - 5424\zeta^2 P^2 + 1356\zeta^2 S^2 \\
& \quad + M_{\min}^2 (-13764D\zeta^3 + 4952D\zeta + 18652\zeta^2 P - 9508P + 4438\zeta^2 S - 198S) \\
& \quad + (14445\zeta^4 - 7710\zeta^2 - 1395) M_{\min}^4 \Big) \Big) \\
& - 2m^2 M_{\min}^2 z^{12} \Big(-30M_{\min}^8 (92D\zeta P + 8D\zeta S + 108\zeta^2 P^2 - 92P^2 - 27\zeta^2 S^2 - 4S^2) \\
& \quad + m^2 M_{\min}^6 \Big(-2004P^2 - 66S^2 + 2004D\zeta P + 132D\zeta S + 2268\zeta^2 P^2 - 567\zeta^2 S^2 \\
& \quad + M_{\min}^2 (3336D\zeta^3 - 508D\zeta - 5588\zeta^2 P + 812P - 542\zeta^2 S + 102S) \Big) \\
& \quad + 2m^4 M_{\min}^4 \Big(832P^2 + 8S^2 - 832D\zeta P - 16D\zeta S - 864\zeta^2 P^2 + 216\zeta^2 S^2 \\
& \quad + M_{\min}^2 (-1518D\zeta^3 + 514D\zeta + 2504\zeta^2 P - 1016P + 266\zeta^2 S - 6S) \\
& \quad + (1155\zeta^4 + 200\zeta^2 - 455) M_{\min}^4 \Big) \\
& \quad + 4m^6 M_{\min}^2 \Big(3456P^2 + 144S^2 - 3456D\zeta P - 288D\zeta S - 4032\zeta^2 P^2 + 1008\zeta^2 S^2 \\
& \quad + M_{\min}^2 (-3078D\zeta^3 - 426D\zeta + 3144\zeta^2 P + 1224P + 1506\zeta^2 S - 186S) \\
& \quad + (1635\zeta^4 - 270\zeta^2 + 115) M_{\min}^4 \Big) \\
& \quad + m^8 \Big(15360P^2 + 1920S^2 - 15360D\zeta P - 3840D\zeta S - 23040\zeta^2 P^2 \\
& \quad + 5760\zeta^2 S^2 + 2304M_{\min}^2 (-3D\zeta^3 - D\zeta + 4\zeta^2 P + 4P + \zeta^2 S - S) \\
& \quad + (2160\zeta^4 + 1440\zeta^2 + 2160) M_{\min}^4 \Big) \Big) \\
& + 2m^2 M_{\min}^4 z^{13} \Big(-9M_{\min}^6 (28D\zeta P + 4D\zeta S + 36\zeta^2 P^2 - 28P^2 - 9\zeta^2 S^2 - 2S^2) \\
& \quad + m^2 M_{\min}^4 \Big(-252P^2 - 18S^2 + 252D\zeta P + 36D\zeta S + 324\zeta^2 P^2 - 81\zeta^2 S^2 \\
& \quad + M_{\min}^2 (420D\zeta^3 - 40D\zeta - 620\zeta^2 P + 20P - 110\zeta^2 S + 30S) \Big) \\
& \quad + m^6 \Big(-96 (16D\zeta P - 2D\zeta S + 12\zeta^2 P^2 - 16P^2 - 3\zeta^2 S^2 + S^2)
\end{aligned}$$

$$\begin{aligned}
& + 120M_{\min}^2 (-3D\zeta^3 - 5D\zeta + 8\zeta^2P + 8P - \zeta^2S + S) + (120\zeta^4 + 400\zeta^2 + 120) M_{\min}^4 \Big) \\
& + m^4 M_{\min}^2 \Big(336P^2 + 24S^2 - 336D\zeta P - 48D\zeta S - 432\zeta^2P^2 + 108\zeta^2S^2 \\
& + M_{\min}^2 (-420D\zeta^3 + 40D\zeta + 620\zeta^2P - 20P + 110\zeta^2S - 30S) + (285\zeta^4 + 110\zeta^2 - 15) M_{\min}^4 \Big) \Big) \\
\hat{c}_U = & M_{\min}^{12} [(5760z - 768) Q_1 - 192Q_2z^2 + 96Q_7z^3 + 80Q_3z^4 - 8Q_4z^5 + 80Q_5z^6 - 4Q_6z^7] \quad (A12) \\
& + 3M_{\min}^{12} z^8 \Big(-98052P^2 - 2658S^2 + 98052D\zeta P + 5316D\zeta S + 108684\zeta^2P^2 - 27171\zeta^2S^2 \\
& + m^2 (8100D\zeta^3 - 12100D\zeta - 10760\zeta^2P + 26360P - 2720\zeta^2S - 1080S) \\
& + (161820\zeta^4 - 171000\zeta^2 + 31260) m^4 \Big) \\
& - 15M_{\min}^{12} z^9 \Big(-12692P^2 - 262S^2 + 12692D\zeta P + 524D\zeta S + 13740\zeta^2P^2 - 3435\zeta^2S^2 \\
& + m^2 (-6396D\zeta^3 + 840D\zeta + 8724\zeta^2P - 1260P + 2034\zeta^2S - 210S) \\
& + (9654\zeta^4 - 9212\zeta^2 + 782) m^4 \Big) \\
& + M_{\min}^{12} z^{10} \Big(-80568P^2 - 1836S^2 + 80568D\zeta P + 3672D\zeta S \\
& + 87912\zeta^2P^2 - 21978\zeta^2S^2 \\
& + m^2 (-21920P - 1320S + 12280D\zeta + 101600\zeta^2P + 14600\zeta^2S - 65400D\zeta^3) \\
& + (13005\zeta^4 - 17910\zeta^2 - 2475) m^4 \Big) \\
& + 5M_{\min}^{12} z^{11} \Big(4392P^2 + 144S^2 - 4392D\zeta P - 288D\zeta S - 4968\zeta^2P^2 \\
& + 1242\zeta^2S^2 + m^2 (4584D\zeta^3 - 800D\zeta - 7576\zeta^2P + 1384P - 796\zeta^2S + 108S) \\
& + (1053\zeta^4 + 630\zeta^2 + 9) m^4 \Big) \\
& + M_{\min}^{12} z^{12} \Big(-3516P^2 - 174S^2 + 3516D\zeta P + 348D\zeta S + 4212\zeta^2P^2 \\
& - 1053\zeta^2S^2 + m^2 (-4596D\zeta^3 + 628D\zeta + 7448\zeta^2P - 872P + 872\zeta^2S - 192S) \\
& + (-2085\zeta^4 - 1090\zeta^2 + 235) m^4 \Big) \\
& + M_{\min}^{12} z^{13} \Big(252P^2 + 18S^2 - 252D\zeta P - 36D\zeta S - 324\zeta^2P^2 + 81\zeta^2S^2 \\
& + m^2 (420D\zeta^3 - 40D\zeta - 620\zeta^2P + 20P - 110\zeta^2S + 30S) + (285\zeta^4 + 110\zeta^2 - 15) m^4 \Big)
\end{aligned}$$

$$\begin{aligned}
\hat{c}_A = & (5760z - 768) M_{\min}^{12} Q_1 - 192M_{\min}^{12} Q_2z^2 \quad (A13) \\
& + 96M_{\min}^{12} z^3 \Big(-3196P^2 - 344S^2 + 3196D\zeta P + 688D\zeta S + 4572\zeta^2P^2 \\
& - 1143\zeta^2S^2 + m^2 (-3180D\zeta^3 + 860D\zeta + 2320\zeta^2P - 1520P + 2020\zeta^2S - 100S) \Big) \\
& + 160M_{\min}^{12} z^4 \Big(2508P^2 + 228S^2 - 2508D\zeta P - 456D\zeta S - 3420\zeta^2P^2 \\
& + 855\zeta^2S^2 + m^2 (4938D\zeta^3 - 1354D\zeta - 3716\zeta^2P + 2396P - 3080\zeta^2S + 156S) \\
& + (594\zeta^4 - 756\zeta^2 + 210) m^4 \Big) \\
& - 32M_{\min}^{12} z^5 \Big(11112P^2 + 804S^2 - 11112D\zeta P - 1608D\zeta S - 14328\zeta^2P^2 + 3582\zeta^2S^2 \\
& + m^2 (18280P + 1095S - 10235D\zeta - 28840\zeta^2P - 21475\zeta^2S + 35895D\zeta^3) \\
& + (13770\zeta^4 - 17220\zeta^2 + 4650) m^4 \Big)
\end{aligned}$$

$$\begin{aligned}
& + 160M_{\min}^{12}z^6 \left(1296P^2 + 72S^2 - 1296D\zeta P - 144D\zeta S - 1584\zeta^2 P^2 \right. \\
& \quad + 396\zeta^2 S^2 + m^2 (6432D\zeta^3 - 1960D\zeta - 5720\zeta^2 P + 3560P - 3572\zeta^2 S + 180S) \\
& \quad \left. + (5271\zeta^4 - 6386\zeta^2 + 1631) m^4 \right) \\
& - 16M_{\min}^{12}z^7 \left(4488P^2 + 192S^2 - 4488D\zeta P - 384D\zeta S - 5256\zeta^2 P^2 + 1314\zeta^2 S^2 \right. \\
& \quad + m^2 (22816P + 876S - 12284D\zeta - 37024\zeta^2 P - 18076\zeta^2 S + 36588D\zeta^3) \\
& \quad \left. + (53655\zeta^4 - 61480\zeta^2 + 14085) m^4 \right) \\
& + z^8 \left(-96M_{\min}^{12} (116D\zeta P + 8D\zeta S + 132\zeta^2 P^2 - 116P^2 - 33\zeta^2 S^2 - 4S^2) \right. \\
& \quad - 320m^2 M_{\min}^{12} (-651D\zeta^3 + 251D\zeta + 754\zeta^2 P - 478P + 274\zeta^2 S - 12S) \\
& \quad + 18480 (27\zeta^4 - 28\zeta^2 + 5) m^4 M_{\min}^{12} \\
& \quad + 16m^8 M_{\min}^4 \left(-5760P^2 - 720S^2 + 5760D\zeta P + 1440D\zeta S + 8640\zeta^2 P^2 \right. \\
& \quad \quad - 2160\zeta^2 S^2 + 960M_{\min}^2 (15D\zeta^3 - 3D\zeta - 12\zeta^2 P + 4P - 9\zeta^2 S + S) \\
& \quad \quad \left. + (-12600\zeta^4 + 10800\zeta^2 - 1080) M_{\min}^4 \right) \Big) \\
& + 80m^2 M_{\min}^4 z^9 \left(2M_{\min}^8 (-267D\zeta^3 + 127D\zeta + 344\zeta^2 P - 248P + 95\zeta^2 S - 3S) \right. \\
& \quad + (-2064\zeta^4 + 1777\zeta^2 - 109) m^2 M_{\min}^8 \\
& \quad + m^6 (2880P^2 + 360S^2 - 2880D\zeta P - 720D\zeta S - 4320\zeta^2 P^2 + 1080\zeta^2 S^2 \\
& \quad \quad + M_{\min}^2 (-6624D\zeta^3 + 1248D\zeta + 5376\zeta^2 P - 1536P + 3936\zeta^2 S - 480S) \\
& \quad \quad \left. + (5580\zeta^4 - 4536\zeta^2 + 396) M_{\min}^4 \right) \Big) \\
& + z^{10} \left(-1280m^2 M_{\min}^{12} (-3D\zeta^3 + 2D\zeta + 4\zeta^2 P - 4P + \zeta^2 S) \right. \\
& \quad + 80 (363\zeta^4 - 218\zeta^2 - 69) m^4 M_{\min}^{12} \\
& \quad - 160m^8 M_{\min}^4 \left(1368P^2 + 144S^2 - 1368D\zeta P - 288D\zeta S - 1944\zeta^2 P^2 \right. \\
& \quad \quad + 486\zeta^2 S^2 + M_{\min}^2 (-2688D\zeta^3 + 448D\zeta + 2336\zeta^2 P - 512P + 1520\zeta^2 S - 192S) \\
& \quad \quad \left. + (2133\zeta^4 - 1518\zeta^2 + 69) M_{\min}^4 \right) \\
& \quad - 16m^{10} M_{\min}^2 \left(9216P^2 + 1152S^2 - 9216D\zeta P - 2304D\zeta S - 13824\zeta^2 P^2 \right. \\
& \quad \quad + 3456\zeta^2 S^2 + 1440M_{\min}^2 (-9D\zeta^3 + D\zeta + 8\zeta^2 P + 5\zeta^2 S - S) \\
& \quad \quad \left. + (7200\zeta^4 - 2880\zeta^2 - 480) M_{\min}^4 \right) \Big) \\
& + z^{11} \left(720 (-3\zeta^4 + \zeta^2 + 2) m^4 M_{\min}^{12} \right. \\
& \quad - 1920m^{10} M_{\min}^2 \left(-96P^2 - 12S^2 + 96D\zeta P + 24D\zeta S + 144\zeta^2 P^2 - 36\zeta^2 S^2 \right. \\
& \quad \quad + M_{\min}^2 (117D\zeta^3 - 9D\zeta - 108\zeta^2 P - 12P - 63\zeta^2 S + 15S) + (-63\zeta^4 + 22\zeta^2 + 1) M_{\min}^4 \Big) \\
& \quad + 80m^8 M_{\min}^4 \left(1224P^2 + 72S^2 - 1224D\zeta P - 144D\zeta S - 1512\zeta^2 P^2 \right. \\
& \quad \quad + 378\zeta^2 S^2 + M_{\min}^2 (-1848D\zeta^3 + 224D\zeta + 1888\zeta^2 P - 256P + 904\zeta^2 S - 96S) \\
& \quad \quad \left. + (1341\zeta^4 - 660\zeta^2 - 69) M_{\min}^4 \right) \Big) \\
& + z^{12} \left(m^{12} \left(-61440P^2 - 7680S^2 + 61440D\zeta P + 15360D\zeta S + 92160\zeta^2 P^2 \right. \right. \\
& \quad \quad \left. \left. - 23040\zeta^2 S^2 - 9216M_{\min}^2 (-3D\zeta^3 - D\zeta + 4\zeta^2 P + 4P + \zeta^2 S - S) \right) \right.
\end{aligned}$$

$$\begin{aligned}
& + (-8640\zeta^4 - 5760\zeta^2 - 8640) M_{\min}^4 \Big) \\
& - 80m^8 M_{\min}^4 \Big(216P^2 - 216D\zeta P - 216\zeta^2 P^2 + 54\zeta^2 S^2 \\
& \quad + M_{\min}^2 (-264D\zeta^3 + 32D\zeta + 352\zeta^2 P - 64P + 88\zeta^2 S) + (171\zeta^4 - 12\zeta^2 - 51) M_{\min}^4 \Big) \\
& - 64m^{10} M_{\min}^2 \Big(1008P^2 + 72S^2 - 1008D\zeta P - 144D\zeta S - 1296\zeta^2 P^2 \\
& \quad + 324\zeta^2 S^2 + M_{\min}^2 (-855D\zeta^3 - 105D\zeta + 870\zeta^2 P + 390P + 420\zeta^2 S - 90S) \\
& \quad + (435\zeta^4 - 70\zeta^2 + 55) M_{\min}^4 \Big) \Big) \\
& + z^{13} \Big(80m^8 M_{\min}^6 (-12D\zeta^3 + 8D\zeta + (9\zeta^4 - 3\zeta^2 - 6) M_{\min}^2 + 16\zeta^2 P - 16P + 4\zeta^2 S) \\
& \quad + m^{10} \Big(-384M_{\min}^2 (16D\zeta P - 2D\zeta S + 12\zeta^2 P^2 - 16P^2 - 3\zeta^2 S^2 + S^2) \\
& \quad + 480M_{\min}^4 (-3D\zeta^3 - 5D\zeta + 8\zeta^2 P + 8P - \zeta^2 S + S) + (480\zeta^4 + 1600\zeta^2 + 480) M_{\min}^6 \Big) \Big)
\end{aligned}$$

$$\begin{aligned}
\hat{c}_B = & (768 - 5760z) M_{\min}^{12} Q_1 + 192M_{\min}^{12} Q_2 z^2 - 96M_{\min}^{12} Q_7 z^3 - 80M_{\min}^{12} Q_3 z^4 \\
& + 8M_{\min}^{12} Q_4 z^5 - 80M_{\min}^{12} Q_5 z^6 + 4M_{\min}^{12} Q_6 z^7 \\
& + z^8 \Big(-9M_{\min}^{12} \Big(-32684P^2 - 886S^2 + 32684D\zeta P + 1772D\zeta S + 36228\zeta^2 P^2 - 9057\zeta^2 S^2 \Big) \\
& \quad + 60m^2 M_{\min}^{12} (-405D\zeta^3 + 605D\zeta + 538\zeta^2 P - 1318P + 136\zeta^2 S + 54S) \\
& \quad - 180 (2697\zeta^4 - 2850\zeta^2 + 521) m^4 M_{\min}^{12} \\
& \quad - 3m^8 M_{\min}^4 \Big(-30720P^2 - 3840S^2 + 30720D\zeta P + 7680D\zeta S \\
& \quad \quad + 46080\zeta^2 P^2 - 11520\zeta^2 S^2 + 5120M_{\min}^2 (15D\zeta^3 - 3D\zeta - 12\zeta^2 P + 4P - 9\zeta^2 S + S) \\
& \quad \quad + (-67200\zeta^4 + 57600\zeta^2 - 5760) M_{\min}^4 \Big) \Big) \\
& + 15M_{\min}^4 z^9 \Big(M_{\min}^8 (12692D\zeta P + 524D\zeta S + 13740\zeta^2 P^2 - 12692P^2 - 3435\zeta^2 S^2 - 262S^2) \\
& \quad + 6m^2 M_{\min}^8 (-1066D\zeta^3 + 140D\zeta + 1454\zeta^2 P - 210P + 339\zeta^2 S - 35S) \\
& \quad + 2 (4827\zeta^4 - 4606\zeta^2 + 391) m^4 M_{\min}^8 \\
& \quad + m^8 \Big(-15360P^2 - 1920S^2 + 15360D\zeta P + 3840D\zeta S + 23040\zeta^2 P^2 - 5760\zeta^2 S^2 \\
& \quad \quad + M_{\min}^2 (8192P + 2560S - 6656D\zeta - 28672\zeta^2 P - 20992\zeta^2 S + 35328D\zeta^3) \\
& \quad \quad + (-29760\zeta^4 + 24192\zeta^2 - 2112) M_{\min}^4 \Big) \Big) \\
& + z^{10} \Big(-54M_{\min}^{12} (1492D\zeta P + 68D\zeta S + 1628\zeta^2 P^2 - 1492P^2 - 407\zeta^2 S^2 - 34S^2) \\
& \quad - 40m^2 M_{\min}^{12} (-1635D\zeta^3 + 307D\zeta + 2540\zeta^2 P - 548P + 365\zeta^2 S - 33S) \\
& \quad - 45 (289\zeta^4 - 398\zeta^2 - 55) m^4 M_{\min}^{12} \\
& \quad + 160m^8 M_{\min}^4 \Big(1368P^2 + 144S^2 - 1368D\zeta P - 288D\zeta S - 1944\zeta^2 P^2 \\
& \quad \quad + 486\zeta^2 S^2 + M_{\min}^2 (-2688D\zeta^3 + 448D\zeta + 2336\zeta^2 P - 512P + 1520\zeta^2 S - 192S) \\
& \quad \quad + (2133\zeta^4 - 1518\zeta^2 + 69) M_{\min}^4 \Big) \\
& \quad + m^{10} M_{\min}^2 \Big(147456P^2 + 18432S^2 - 147456D\zeta P - 36864D\zeta S \\
& \quad \quad - 221184\zeta^2 P^2 + 55296\zeta^2 S^2 + 23040M_{\min}^2 (-9D\zeta^3 + D\zeta + 8\zeta^2 P + 5\zeta^2 S - S) \\
& \quad \quad + (115200\zeta^4 - 46080\zeta^2 - 7680) M_{\min}^4 \Big) \Big) \\
& + z^{11} \Big(90M_{\min}^{12} (244D\zeta P + 16D\zeta S + 276\zeta^2 P^2 - 244P^2 - 69\zeta^2 S^2 - 8S^2)
\end{aligned} \tag{A14}$$

$$\begin{aligned}
& + 20m^2 M_{\min}^{12} (-1146D\zeta^3 + 200D\zeta + 1894\zeta^2 P - 346P + 199\zeta^2 S - 27S) \\
& - 45 (117\zeta^4 + 70\zeta^2 + 1) m^4 M_{\min}^{12} \\
& + 1920m^{10} M_{\min}^2 \left(-96P^2 - 12S^2 + 96D\zeta P + 24D\zeta S + 144\zeta^2 P^2 - 36\zeta^2 S^2 \right. \\
& \quad \left. + M_{\min}^2 (117D\zeta^3 - 9D\zeta - 108\zeta^2 P - 12P - 63\zeta^2 S + 15S) + (-63\zeta^4 + 22\zeta^2 + 1) M_{\min}^4 \right) \\
& - 80m^8 M_{\min}^4 \left(1224P^2 + 72S^2 - 1224D\zeta P - 144D\zeta S - 1512\zeta^2 P^2 \right. \\
& \quad \left. + 378\zeta^2 S^2 + M_{\min}^2 (-1848D\zeta^3 + 224D\zeta + 1888\zeta^2 P - 256P + 904\zeta^2 S - 96S) \right. \\
& \quad \left. + (1341\zeta^4 - 660\zeta^2 - 69) M_{\min}^4 \right) \\
& + z^{12} \left(-3M_{\min}^{12} (1172D\zeta P + 116D\zeta S + 1404\zeta^2 P^2 - 1172P^2 - 351\zeta^2 S^2 - 58S^2) \right. \\
& \quad - 4m^2 M_{\min}^{12} (-1149D\zeta^3 + 157D\zeta + 1862\zeta^2 P - 218P + 218\zeta^2 S - 48S) \\
& \quad + 5 (417\zeta^4 + 218\zeta^2 - 47) m^4 M_{\min}^{12} \\
& \quad + 80m^8 M_{\min}^4 \left(216P^2 - 216D\zeta P - 216\zeta^2 P^2 + 54\zeta^2 S^2 \right. \\
& \quad \left. + M_{\min}^2 (-264D\zeta^3 + 32D\zeta + 352\zeta^2 P - 64P + 88\zeta^2 S) + (171\zeta^4 - 12\zeta^2 - 51) M_{\min}^4 \right) \\
& \quad + 64m^{10} M_{\min}^2 \left(1008P^2 + 72S^2 - 1008D\zeta P - 144D\zeta S - 1296\zeta^2 P^2 \right. \\
& \quad \left. + 324\zeta^2 S^2 + M_{\min}^2 (-855D\zeta^3 - 105D\zeta + 870\zeta^2 P + 390P + 420\zeta^2 S - 90S) \right. \\
& \quad \left. + (435\zeta^4 - 70\zeta^2 + 55) M_{\min}^4 \right) \\
& \quad + m^{12} \left(61440P^2 + 7680S^2 - 61440D\zeta P - 15360D\zeta S - 92160\zeta^2 P^2 \right. \\
& \quad \left. + 23040\zeta^2 S^2 + 9216M_{\min}^2 (-3D\zeta^3 - D\zeta + 4\zeta^2 P + 4P + \zeta^2 S - S) \right. \\
& \quad \left. + (8640\zeta^4 + 5760\zeta^2 + 8640) M_{\min}^4 \right) \\
& + z^{13} \left(9M_{\min}^{12} (28D\zeta P + 4D\zeta S + 36\zeta^2 P^2 - 28P^2 - 9\zeta^2 S^2 - 2S^2) \right. \\
& \quad + 10m^2 M_{\min}^{12} (-42D\zeta^3 + 4D\zeta + 62\zeta^2 P - 2P + 11\zeta^2 S - 3S) - 5 (57\zeta^4 + 22\zeta^2 - 3) m^4 M_{\min}^{12} \\
& \quad + m^{10} M_{\min}^2 \left(384 (16D\zeta P - 2D\zeta S + 12\zeta^2 P^2 - 16P^2 - 3\zeta^2 S^2 + S^2) \right. \\
& \quad \left. - 480M_{\min}^2 (-3D\zeta^3 - 5D\zeta + 8\zeta^2 P + 8P - \zeta^2 S + S) + (-480\zeta^4 - 1600\zeta^2 - 480) M_{\min}^4 \right) \\
& \quad \left. - 80m^8 M_{\min}^6 (-12D\zeta^3 + 8D\zeta + (9\zeta^4 - 3\zeta^2 - 6) M_{\min}^2 + 16\zeta^2 P - 16P + 4\zeta^2 S) \right),
\end{aligned}$$

where we have introduced the shorthand notations $Q_i(\zeta)$ for polynomials of the variable ζ that are invariant under the permutation of the two quarkonia:

$$Q_1 = \left(-8P^2 - S^2 + 8D\zeta P + 2D\zeta S + 12\zeta^2 P^2 - 3\zeta^2 S^2 \right), \quad (\text{A15})$$

$$\begin{aligned}
Q_2 = \left(-128m^2 P - 804P^2 - 8m^2 S - 96S^2 + 72D\zeta m^2 + 804D\zeta P + 192D\zeta S \right. \\
\left. + 192\zeta^2 m^2 P + 1188\zeta^2 P^2 + 168\zeta^2 m^2 S - 297\zeta^2 S^2 - 264D\zeta^3 m^2 \right), \quad (\text{A16})
\end{aligned}$$

$$\begin{aligned}
Q_3 = \left(420m^4 + 4792m^2 P + 4392P^2 + 312m^2 S + 387S^2 - 2708D\zeta m^2 \right. \\
\left. - 4392D\zeta P - 774D\zeta S - 1512\zeta^2 m^4 - 7432\zeta^2 m^2 P - 5940\zeta^2 P^2 - 6160\zeta^2 m^2 S \right. \\
\left. + 1485\zeta^2 S^2 + 9876D\zeta^3 m^2 + 1188\zeta^4 m^4 \right), \quad (\text{A17})
\end{aligned}$$

$$Q_4 = \left(18600m^4 + 72640m^2 P + 25812P^2 + 4260m^2 S + 1539S^2 \right)$$

$$\begin{aligned}
& -40580D\zeta m^2 - 25812D\zeta P - 3078D\zeta S - 68880\zeta^2 m^4 - 113920\zeta^2 m^2 P \\
& - 31968\zeta^2 P^2 - 84820\zeta^2 m^2 S + 7992\zeta^2 S^2 + 141780D\zeta^3 m^2 + 55080\zeta^4 m^4),
\end{aligned} \tag{A18}$$

$$\begin{aligned}
Q_5 = & \left(3262m^4 + 6844m^2 P - 828P^2 + 306m^2 S - 81S^2 - 3728D\zeta m^2 \right. \\
& + 828D\zeta P + 162D\zeta S - 12772\zeta^2 m^4 - 10660\zeta^2 m^2 P + 1152\zeta^2 P^2 - 6634\zeta^2 m^2 S \\
& \left. - 288\zeta^2 S^2 + 11964D\zeta^3 m^2 + 10542\zeta^4 m^4 \right),
\end{aligned} \tag{A19}$$

$$\begin{aligned}
Q_6 = & \left(56400m^4 + 77636m^2 P - 67932P^2 + 1566m^2 S - 2943S^2 \right. \\
& - 40384D\zeta m^2 + 67932D\zeta P + 5886D\zeta S - 245560\zeta^2 m^4 - 110684\zeta^2 m^2 P \\
& \left. + 79704\zeta^2 P^2 - 52646\zeta^2 m^2 S - 19926\zeta^2 S^2 + 107988D\zeta^3 m^2 + 213720\zeta^4 m^4 \right),
\end{aligned} \tag{A20}$$

$$\begin{aligned}
Q_7 = & \left(-1520m^2 P - 3116P^2 - 100m^2 S - 334S^2 + 860D\zeta m^2 + 3116D\zeta P \right. \\
& \left. + 668D\zeta S + 2320\zeta^2 m^2 P + 4452\zeta^2 P^2 + 2020\zeta^2 m^2 S - 1113\zeta^2 S^2 - 3180D\zeta^3 m^2 \right).
\end{aligned} \tag{A21}$$

Appendix B: Diquarkonium fragmentation function in the helicity basis

In this appendix, we provide the technical details of the evaluation of the full set of Feynman diagrams shown in Fig. 2, which is needed to obtain the dihadron fragmentation function of the charm quark. This evaluation is highly nontrivial because of the large number of combinatorial pairwise combinations arising from the interference between different diagrams in the amplitude and its conjugate, and the necessity of evaluating traces containing up to twenty Dirac matrices, which typically results in extremely lengthy analytical expressions. Fortunately, this evaluation simplifies significantly in the helicity basis. Since the initiating (parent) and final-state quarks can only have two helicity states, we only need to determine the amplitudes with and without a helicity flip of the active quark. Precisely, the squared amplitude appearing in the integrand of Eq. (47) can be represented as

$$\sum_{i,j} |\mathcal{M}_{\text{frag}}^{ji}(k_1, P_1, P_2; K)|^2 = \frac{\text{Tr} \left[\gamma_+ (\not{K} + m) \hat{d}^\dagger (\not{K} - \not{P}_1 - \not{P}_2 + m) \hat{d} (\not{K} + m) \right]}{(K^2 - m^2 + i0)^2} \tag{B1}$$

where $\hat{d}$ denotes the sum of the contributions of the cut diagrams shown in Fig. 4. The expressions for $\hat{d}$ and $\hat{d}^\dagger$ also include the polarization vectors of the produced quarkonia; for the unpolarized case, we sum over these polarizations using the identity (A7). Since the direct evaluation of the trace on the right-hand side of Eq. (B1) is mathematically challenging, we switch to the helicity representation. Using the fact that the final-state quark momentum $K - P_1 - P_2$ is on-shell, we may write

$$(\not{K} - \not{P}_1 - \not{P}_2 + m) = \begin{cases} \sum_{h_2} u_{h_2}(K - P_1 - P_2) \bar{u}_{h_2}(K - P_1 - P_2), & K^+ > P_1^+ + P_2^+ \\ -\sum_{h_2} v_{h_2}(K - P_1 - P_2) \bar{v}_{h_2}(K - P_1 - P_2), & K^+ < P_1^+ + P_2^+ \end{cases}. \tag{B2}$$

Although the momentum K is off-shell ($K^2 \neq m^2$), we can decompose the numerator $(\not{K} + m)$ into a sum over on-shell spinors and an additional term that accounts for the off-shellness of the quark:

$$(\not{K} + m) = \left(K^+ \gamma^- + \frac{\mathbf{K}_\perp^2 + m^2}{2K^+} \gamma^+ - \mathbf{K}_\perp \cdot \boldsymbol{\gamma}_\perp + m \right) + \gamma^+ \left(K^- - \frac{\mathbf{K}_\perp^2 + m^2}{2K^+} \right) = \tag{B3}$$

$$\begin{aligned}
& = \theta(k'^+) \sum_{h_1} u_{h_1}(K') \bar{u}_{h_1}(K') - \theta(-K^+) \sum_{h_1} v_{h_1}(K') \bar{v}_{h_1}(K') + \gamma^+ \left(K^- - \frac{\mathbf{K}_\perp^2 + m^2}{2K^+} \right) \\
& K' = \left(K^+, \frac{\mathbf{K}_\perp^2 + m^2}{2K^+}, \mathbf{K}_\perp \right).
\end{aligned} \tag{B4}$$

In the framework of light-cone perturbation theory [19, 88], the term proportional to γ^+ corresponds to the instantaneous fermion propagator. Since $(\not{K} + m)$ in Eq. (B1) is contracted with γ^+ , the identity $\gamma^+\gamma^+ = 0$ ensures that this instantaneous contribution vanishes. Furthermore, a cyclic permutation of the matrices within the trace, combined with the identities (40) and (41), allows us to express the squared amplitude of the $q \rightarrow q\mathcal{Q}_1\mathcal{Q}_2$ subprocess as a sum of helicity-flip and helicity-conserving contributions,

$$\sum_{i,j} |\mathcal{M}_{\text{frag}}^{\text{ji}}(k_1, P_1, P_2; K)|^2 = 2K^+ \sum_{h_1} \mathcal{M}_{h_1, h_1}^* \mathcal{M}_{h_1, h_1} + 2K^+ \sum_{h_1} \mathcal{M}_{h_1, -h_1}^* \mathcal{M}_{-h_1, h_1}. \quad (\text{B5})$$

Using the algebraic identity

$$\begin{aligned} & |\mathcal{M}_{++}|^2 + |\mathcal{M}_{--}|^2 + |\mathcal{M}_{+-}|^2 + |\mathcal{M}_{-+}|^2 = \\ & = 2 \left[\left(\frac{\mathcal{M}_{++} + \mathcal{M}_{--}}{2} \right)^2 + \left(\frac{\mathcal{M}_{++} - \mathcal{M}_{--}}{2} \right)^2 + \left(\frac{\mathcal{M}_{+-} + \mathcal{M}_{-+}}{2} \right)^2 + \left(\frac{\mathcal{M}_{+-} - \mathcal{M}_{-+}}{2} \right)^2 \right], \end{aligned} \quad (\text{B6})$$

and the properties of the light-cone Dirac spinors [19, 88]:

$$\begin{aligned} \bar{u}_h(k_1) \gamma^+ u_{h'}(k_2) &= 2\sqrt{k_1^+ k_2^+} \delta_{h, h'}, & \bar{u}_h(k_1) \gamma_5 \gamma^+ u_{h'}(k_2) &= 2\sqrt{k_1^+ k_2^+} \delta_{h, h'} (-1)^{h+1/2}, \\ \bar{u}_h(k_1) i\gamma_2 \gamma^+ u_{h'}(k_2) &= 2\sqrt{k_1^+ k_2^+} \delta_{h, -h'}, & \bar{u}_h(k_1) i\gamma_5 \gamma_2 \gamma^+ u_{h'}(k_2) &= 2\sqrt{k_1^+ k_2^+} \delta_{h, -h'} (-1)^{h+1/2}, \end{aligned}$$

we obtain

$$\mathcal{M}_{++} + \mathcal{M}_{--} = \frac{\text{Tr} \left[\hat{d} (\not{K} + m) \gamma^+ (\not{K} - \not{P}_1 - \not{P}_2 + m) \right]}{2\sqrt{K^+ (K^+ - P_1^+ - P_2^+)}} \quad (\text{B7})$$

$$\mathcal{M}_{++} - \mathcal{M}_{--} = \frac{\text{Tr} \left[\hat{d} (\not{K} + m) \gamma_5 \gamma^+ (\not{K} - \not{P}_1 - \not{P}_2 + m) \right]}{2\sqrt{K^+ (K^+ - P_1^+ - P_2^+)}} \quad (\text{B8})$$

$$\mathcal{M}_{+-} + \mathcal{M}_{-+} = \frac{\text{Tr} \left[\hat{d} (\not{K} + m) i\gamma_2 \gamma^+ (\not{K} - \not{P}_1 - \not{P}_2 + m) \right]}{2\sqrt{K^+ (K^+ - P_1^+ - P_2^+)}} \quad (\text{B9})$$

$$\mathcal{M}_{+-} - \mathcal{M}_{-+} = \frac{\text{Tr} \left[\hat{d} (\not{K} + m) i\gamma_5 \gamma_2 \gamma^+ (\not{K} - \not{P}_1 - \not{P}_2 + m) \right]}{2\sqrt{K^+ (K^+ - P_1^+ - P_2^+)}} \quad (\text{B10})$$

The contributions (B8)–(B10) involve combinations that are antisymmetric with respect to the permutation of the two quarkonia, and for this reason are numerically suppressed in the $\mathbf{R}_\perp$ -integrated collinear fragmentation functions. While the following evaluation is straightforward, the explicit *FeynCalc*-generated expressions for the contribution of charm to the dihadron fragmentation function are very lengthy and can't be presented here due to space limitations. The full expressions can be provided on demand as a C/C++ code.

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- [90] As can be seen from the analytic results in Ref. [36], a similar $1/z$ pole is present in front of all color-octet LDMEs, but the residue is proportional to $\ln(\mu^2/4m_c^2)$, causing the pole to vanish for $\mu = 2m_c$.
- [91] For the dominant color singlet LDMEs, the amplitude $\mathcal{M}_{\text{frag}}^{c',c}$ also belongs to the color singlet irreducible representations (namely, is diagonal w.r.t. color singlet indices). However, for color octet LDMEs the amplitude may include nondiagonal contributions $\sim (t^a)_{c',c}$.
- [92] Due to C -parity of strong interactions, a similar relation holds for the conjugate matrix projected onto spinors of antiquarks $v, \bar{v}$.
- [93] Indeed, in heavy-quark pair production, the produced active quark is on-shell and contributes a phase-space factor $dk'^+ d^2\mathbf{k}'_{\perp}/[2k'^+(2\pi)^3]$, whereas in the production of a heavy quark and a quarkonium pair, the active quark contributes with $dk_1^+ d^2\mathbf{k}_{1\perp}/[2k_1^+(2\pi)^3]$. The transverse momenta $\mathbf{k}_{1\perp}$ and $\mathbf{k}'_{\perp}$ are related by Eq. (33), so that $d^2\mathbf{k}_{1\perp} = d^2\mathbf{k}'_{\perp}$. The longitudinal components are related by $k_1^+ = (1-z)k'^+$, and the factor of z in the denominator comes from the definition (29).